

Pristine Shine
Desktop Application



*A project submitted
In partial fulfillment of requirements for the degree of
Bachelor of Science in Software Engineering*

By

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Starting with the name of Allah, the Most Merciful and the Most Beneficent.

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UNDERTAKING

This is to certify that the project entitled “Pristine Shine – Car Detailing Management System” is an original and authentic effort completed by the undersigned group members as a part of the degree requirement for BSSE (Bachelor of Science in Software Engineering) at the Department of Computer Science, NFC Institute of Engineering and Technology, Multan, Pakistan.

All phases of the project including requirement analysis, system design, development, and implementation have been carried out solely by the group members. This project reflects our collective contribution and sincere effort over the course of the academic term.

We also affirm that this project is a unique and original work, has not been copied or plagiarized, and has not been submitted to any other institution for academic or professional purposes.

ABSTRACT

Pristine Shine is an intelligent car detailing studio management system designed to streamline daily operations such as client management, appointment booking, service tracking, financial reporting, and AI-based damage detection. This Windows-based desktop application was developed using C# (.NET Framework), MySQL for database storage, and Python-based AI modules for image recognition. The system features separate portals for Admin and Employees, ensuring role-based access control. Admins can manage users, services, and view charts and income reports, while employees handle bookings, search clients by CNIC or car number, and update service statuses. Integrated with Mocean SMS API, the system can send automated service reminders to clients. The AI module detects car damage via image input to assist employees during check-ins. By automating and digitizing core functions, Pristine Shine improves operational efficiency, reduces manual errors, and enhances customer engagement in car care businesses.



PRISTINE SHINE

YOUR CAR, OUR PERFECTION.

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Chapter 1 Introduction

Introduction

In the rapidly evolving landscape of automotive services, efficiency, precision, and customer satisfaction are key to business success. Car detailing studios, in particular, require careful coordination of appointments, client data, vehicle history, service records, and financial tracking. However, many such businesses still rely on manual systems or basic software, which often leads to inefficiencies, missed opportunities, and human error. This project aims to develop a comprehensive desktop application for Pristine Shine, a professional car detailing studio, to automate and streamline its daily operations. The system is designed to support both administrative and employee roles, offering features such as client and vehicle management, appointment booking, service tracking, financial reporting, and automated customer reminders. By integrating modern software technologies and intelligent automation, this project seeks to enhance business performance, reduce manual workload, and deliver a better experience for both staff and customers.

1.1 Domain

Business Process Management in Automotive Services

- Focuses on automating and optimizing workflows in the automotive service industry.

1. Sub-Domain: Car Detailing Operations Management

- Addresses the specific needs of car detailing businesses, such as managing services, client interactions, employee tasks, and financial records.

1.2 Problem Statement

Car detailing studios face several operational challenges that hinder efficiency and customer satisfaction, including:

1. **Manual Scheduling:** Difficulty in tracking scheduled services and following up with clients for maintenance or repeat services.
2. **Task Assignment Issues:** Inefficient assignment of tasks to employees, often leading to mismanagement and delays.
3. **Financial Tracking:** Lack of an accurate and transparent system for monitoring income and expenses.
4. **Service Recommendations:** Relying solely on manual inspection to determine the required services, increasing the potential for errors.
5. **Client Data Management:** Limited tools to maintain and organize customer records, resulting in a lack of personalized engagement.

Pristine Shine solves these problems by automating processes, integrating AI tools, and offering a centralized platform for managing all aspects of a car detailing studio.

1.3 Motivation

The motivation behind developing **Pristine Shine** stems from the need to:

1. **Increase Efficiency:** Replace manual processes with automation to save time and reduce errors.

2. **Adopt Modern Technology:** Leverage AI to improve service recommendations and decision-making.
3. **Enhance Customer Retention:** Provide timely service reminders and a superior customer experience to encourage repeat business.
4. **Support Employee Productivity:** Offer tools for easy task assignment and management to ensure smooth operations.
5. **Meet Industry Trends:** Align with the growing demand for digital solutions in the automotive service sector.

1.4 Definition of Terms

Sr No	Terms	Meaning
1	Admin/Owner	The user role with access to advanced features like financial tracking, employee management, and operational oversight.
2	Employee	The user role focused on daily service operations, including appointment booking and task completion.
3	AI Detection Tool	A feature that uses artificial intelligence to analyze car images and recommend suitable services.
4	Schedule Service	Services that are due for follow-up, such as routine maintenance or checkups after a predefined interval.
5	Financial Report	A module for tracking studio income and expenses, aiding profitability analysis.

Table 1 Definition of term

1.5 Workflow

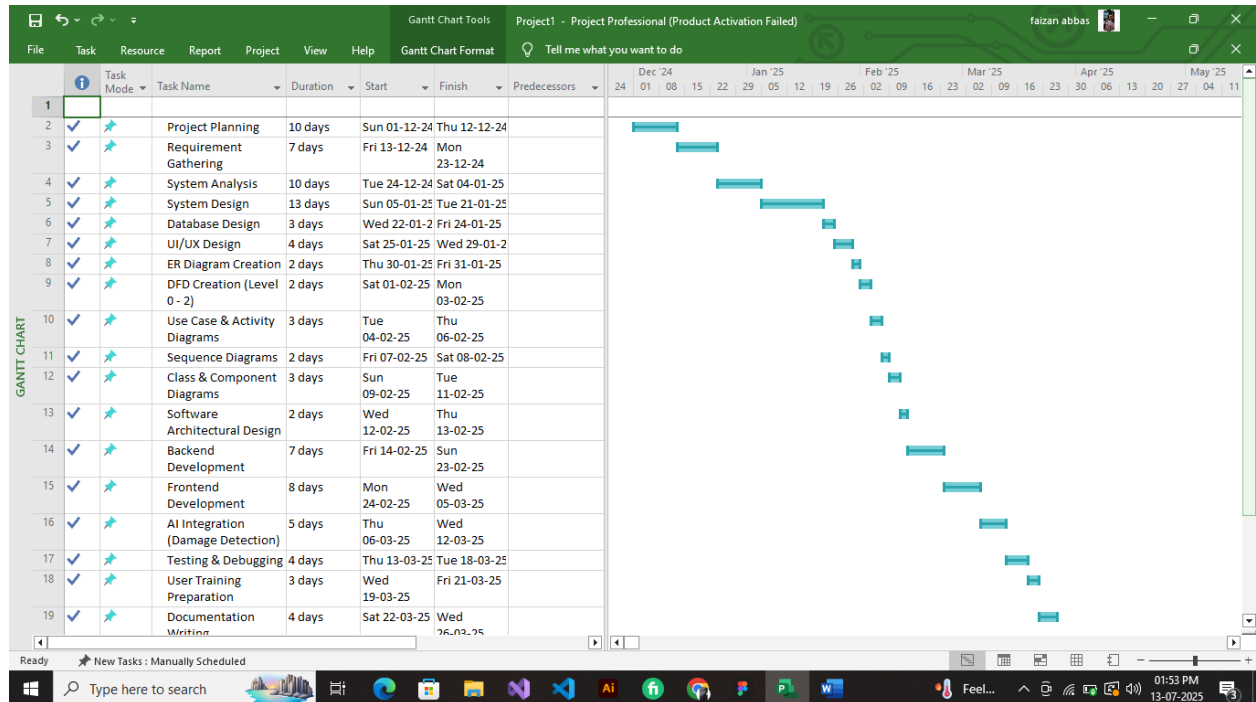


Figure 1 MS Project screen shot

1.6 Goals of the Project

The goals of **Pristine Shine** include:

- **Automating Studio Operations:** Reduce manual effort by automating appointment booking, task assignment, and service reminders.
- **Enhancing Customer Satisfaction:** Provide accurate service recommendations through AI and maintain personalized client engagement.
- **Streamlining Task Management:** Ensure efficient employee task allocation and monitoring through a dedicated schedule page.
- **Centralizing Data:** Offer a unified platform for managing client details, car records, employee data, and financial information.
- **Ensuring Security:** Use role-based access controls to protect sensitive data and prevent unauthorized access.
- **Improving Decision-Making:** Offer insights into financial performance and operational metrics to aid managerial decisions. The process of reserving a service slot for a client's car.

Chapter 2 Existing System

Existing System

The car detailing industry, while growing rapidly, often lags in adopting efficient and specialized technological solutions. Many businesses still rely on outdated, manual methods such as paper-based logs or spreadsheets for managing appointments, tracking customer history, and generating invoices. These approaches are time-consuming, prone to errors, and limit the ability of businesses to scale efficiently. Existing software solutions available in the market, though helpful to some extent, are often generic or designed for broader service industries, making them less effective for the specific needs of car detailing studios. While some car-detailing-specific tools do exist, they are typically expensive, dependent on constant internet connectivity, and lack cutting-edge features like AI-powered recommendations or tailored financial reporting. These limitations highlight the need for a dedicated, user-friendly, and cost-effective system that caters exclusively to the unique requirements of car detailing businesses, a gap that *Pristine Shine* is designed to fill.

2.1 Relative Work

In the automotive detailing industry, various software systems are available to assist car detailing businesses with managing appointments, customer data, billing, and inventory. These systems typically fall into three categories:

1. Manual Systems:

- **Description:**
 - Many businesses still rely on manual systems like Excel sheets, appointment books, and physical records.
- **Limitations:**
 - Time-consuming processes.
 - High chance of human error (e.g., missed appointments, incorrect billing).
 - No automation for reminders or service recommendations.

2. Generic Business Management Software:

- **Description:**
 - Some car detailing businesses use generic solutions like QuickBooks, Zoho CRM, or Microsoft Excel for handling financial records and customer interactions.
- **Limitations:**
 - Lack of domain-specific features tailored to the car detailing industry.
 - Requires significant customization and integration with other tools to fit business needs.

3. Car Detailing-Specific Software:

- **Description:**

Several software solutions are explicitly designed for car detailing, such as Auto Detailer Pro, Shopmonkey, and Detail Bookie. These tools offer features like appointment scheduling, inventory management, and customer notifications.

- **Limitations:**

- Most of these solutions are web-based or SaaS platforms, requiring a stable internet connection.
- High subscription costs, making them unsuitable for small businesses.
- Limited integration of AI-based recommendations for service optimization.

2.2 Existing System and their features

Software Name	Description	Features	Limitations	Link
Shopmonkey	A modern auto shop management platform for scheduling, invoicing, and customer tracking.	Scheduling, inventory management, invoicing, customer tracking.	High monthly subscription fee. Requires internet.	https://www.shopmonkey.io
Detail Bookie	An all-in-one solution for auto detailers to manage bookings and payments.	Online booking, task management, payment processing.	No AI-based service recommendations. Lacks offline functionality.	https://www.detailbookie.com
Auto Detailer Pro	Lightweight software for small detailing businesses focusing on customer and service tracking.	Client database, appointment reminders, service history tracking.	Limited to basic functionality. No integration with advanced analytics.	https://www.autodetailerpro.com
QuickBooks	Leading financial management tool with support for small business invoicing and reporting.	Financial management, invoicing, and basic customer management.	Requires customization. Lacks specific car detailing features.	https://quickbooks.intuit.com
Zoho CRM	CRM software for tracking customer	Customer relationship management, marketing	No car-specific modules. Complex to set up	https://www.zoho.com/crm

	interactions and automating marketing/sales.	automation, and sales tracking.	and use for car detailing.	
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Table 2 Other Software in Market

2.3 Novelty of *Pristine Shine*

Pristine Shine introduces innovative features tailored specifically for car detailing studios, bridging the gaps left by existing systems:

1. AI-Powered Detection:

- Unlike other systems, *Pristine Shine* incorporates an AI-based car image analysis tool that recommends services based on the uploaded images. This ensures accurate and data-driven suggestions for customers.

2. Offline Desktop Solution:

- Most car detailing software requires an internet connection to function. *Pristine Shine*, being a desktop application, works offline, eliminating dependency on internet availability.

3. Integrated Scheduling and Notification System:

- With a built-in Twilio integration, administrators can send SMS reminders for due services directly from the application.
- Employees can assign and track tasks, ensuring operational efficiency.

4. Tailored Financial Reporting:

- Unlike generic tools, *Pristine Shine* provides financial reports designed explicitly for car detailing businesses, tracking income, expenses, and service trends.

5. User-Friendly Interface:

- A simple and intuitive interface ensures that employees with minimal technical knowledge can operate the system efficiently.

2.4 Solutions Available in *Pristine Shine*

Pristine Shine offers several features that address the pain points of existing systems:

1. For Administrators:

- Comprehensive **dashboard** to monitor business operations at a glance.
- Ability to **search cars** and view their service history instantly.
- Automated reminders for **due services**, sent to customers via SMS.
- Detailed **financial reports** showing income and expenses.
- Management of employee tasks and schedules.

2. For Employees:

- Seamless **appointment booking** with built-in payment processing and bill generation.
- A dynamic **schedule page** to assign tasks and track completed work.
- **AI detection page** to analyze car images and recommend services.
- Simplified **car search** functionality.

3. For Clients:

- Timely notifications for due services.
- Clear, professional bills and invoices.

2.5 Software Requirements Specification (SRS)

2.5.1 User Interfaces

These are the main interfaces through which users interact with the Pristine Shine system:

Interface	User Role	Purpose
Login Interface	Admin, Employee	Secure access for authorized users. Session is initiated here.
Admin Dashboard	Admin	Access to user management, services, reports, and system settings.
Employee Dashboard	Employee	Access to appointments, car search, damage detection, and service management.
Search Interface	Employee	Search for cars or clients using CNIC or car number.
Appointment Management Page	Employee	Book, edit, and complete appointments.
AI Damage Detection Page	Employee	Upload car images for damage detection using AI model.
Reminder Notification Interface	Admin	Send SMS reminders via Mocean API to clients with pending appointments.
Financial Report Page	Admin	View income and expenses with chart support (Guna Charts).

Password Recovery Interface	Admin/Employee	Forgot password feature with code verification and reset.
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Table 3 User Interfaces

2.5.2 Hardware Interfaces

Component	Minimum Requirement
Processor	Intel Core i3 or higher
RAM	4 GB minimum (8 GB recommended)
Storage	500 MB free disk space (excluding MySQL and Python dependencies)
Network	Required only for Mocean SMS API or updates (optional)

Table 4 Hardware Interfaces

2.5.3 Software Interfaces

Component	Details
Operating System	Windows 10 or above (64-bit)
.NET Framework	Version 4.8 (for WinForms app)
Database	MySQL Server 8.0 (Localhost)
Python Runtime	Python 3.11.5 (for AI Model execution)
Mocean API	Used for sending SMS reminders to clients
Guna UI2 / Guna Charts	For enhanced WinForms UI and data visualization
OpenAI / TensorFlow (via Python)	Required for AI-based damage detection module
Inno Setup	Used for bundling offline installer with auto-config scripts

Table 5 Software Interfaces

2.6 Functional Requirement

Functional Requirement		Actor(s)	Description
1.	User Authentication	Admin, Employee	The system must allow admins and employees to log in securely using registered credentials (email, password). Admins and employees must be able to reset their password.
2.	View Dashboard	Admin	The admin should be able to view a system dashboard showing business statistics like income, services done, total employees, and profits.
3.	Appointment Management	Employee	Employees can book appointments, including car details, services, employee assignment, and appointment date. The employee will see the total, discount, grand total, and payment method (cash or online).
4.	Service Management	Admin	Admin can manage the list of services, including adding, editing, or deleting services. Each service has a name and price.
5.	Schedule Management	Admin, Employee	Admins can see due services and manage the schedule. Employees can view their assigned tasks and mark them as completed.
6.	View Appointment Details	Admin	Admin should be able to view details of all appointments, including client and car details, services, and employee assignments.
7.	Client Management	Admin	Admin should be able to manage client data, including adding, editing, and deleting client records.
8.	Income and Expense Management	Admin	Admin should be able to view and manage income and expense records. This includes adding income or expenses and tracking financials to calculate monthly profit.
9.	Employee Management	Admin	Admin can manage employee information, including adding, editing, or removing employees, as well as their roles and salary details.
10.	Notification Management	Admin	Admin can send notifications or alerts to clients regarding due services.
11.	Service Completion	Employee	Employees can mark services as completed once the task is done, updating the service status.
12.	Appointment Search	Admin, Employee	Admin and employees can search for an appointment using car number to view previous appointments.

13.	Appointment Notification	Employee	Employees will receive a notification about their assigned work once an appointment is booked.
14.	Appointment Status Update	Employee	Employees can mark tasks as "done" to update the appointment status and remove it from the schedule view.

Table 6 Functional Requirements

2.7 Non-Functional Requirements

No.	Non-Functional Requirement	Description
1	Performance	The system must support up to 25 concurrent users without significant performance degradation.
2	Reliability	The system should maintain an uptime of at least 95% to ensure service availability.
3	Response Time	The system should respond to any user query or action within 5 seconds under normal load.
4	Usability	The interface should be easy to navigate and require minimal training for new employees.
5	Compatibility	The system must run on Windows 10 and above, with 4 GB of RAM and 2 GB of storage.

Table 7 Non-Functional Requirements

2.8 Use Case Diagrams

Admin use case

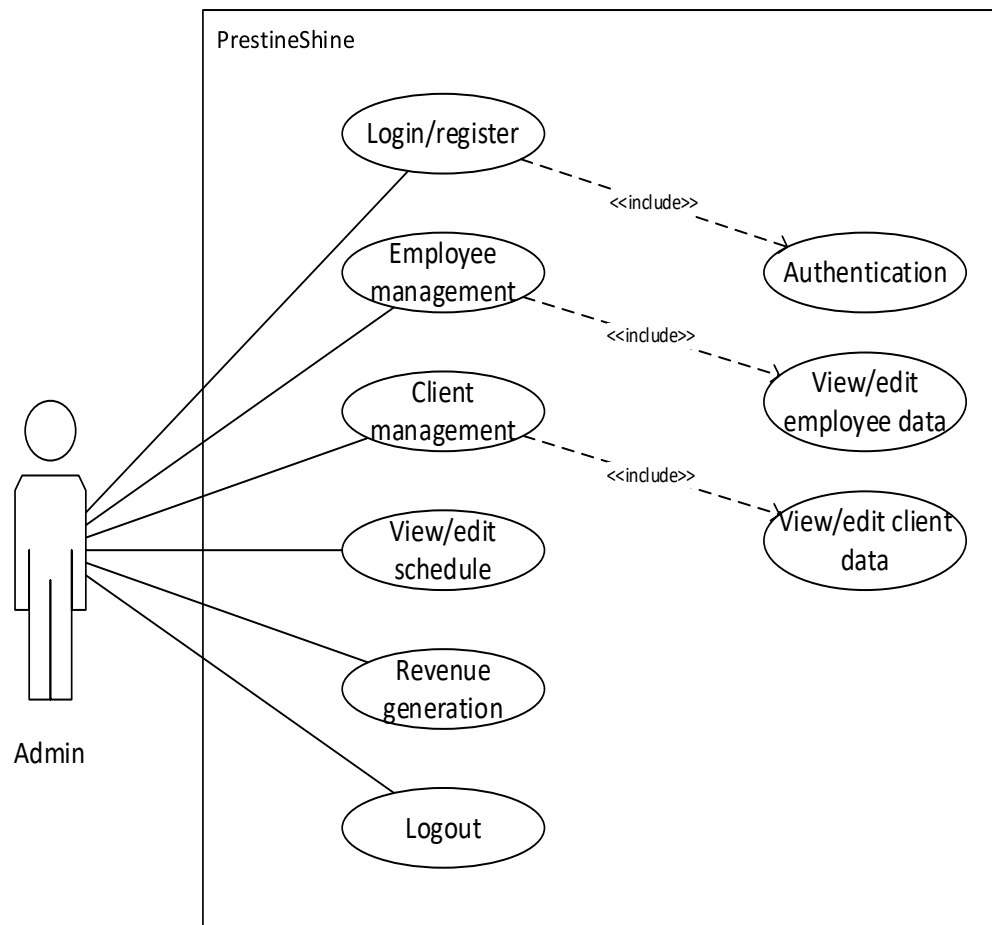


Figure 2 Admin usecase

Employee use case

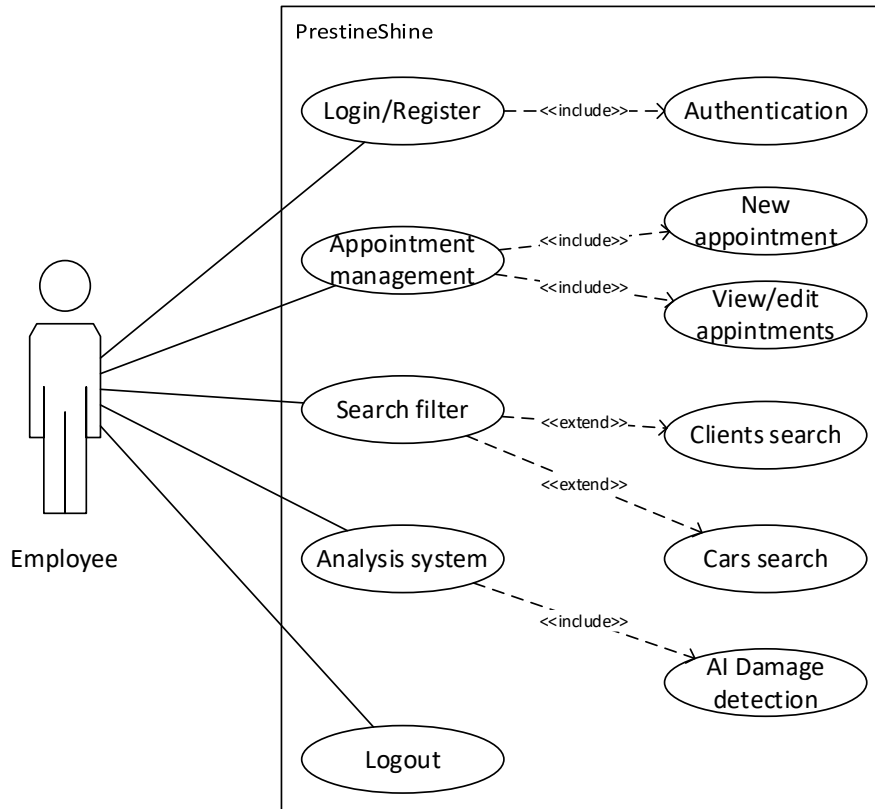


Figure 3 Employee usecase

2.9 Requirement Shell

2.9.1 Functional Requirement

Requirement # 1	Requirement Type: FR	Event/Use Case: Appointment Creation
Description: The system allows employees to create and manage appointments by entering car details, services, and employee assignments.		
Rationale: This feature is essential for managing daily operations efficiently.		
Originator: Stakeholder		
Fit Criterion: Appointment is created successfully, and details are stored.		
Priority: High		
Customer Satisfaction: 5		Customer Dissatisfaction: 2
Supporting Material: None		Conflict: None
History: 14/12/24		

Requirement # 2	Requirement Type: FR	Event/Use Case: Appointment Payment
Description: System allows employees to process payments for appointments with options for cash or online payments.		
Rationale: Enables seamless transaction processing.		
Originator: Stakeholder		
Fit Criterion: Payment is processed and logged correctly in the system.		
Priority: High		
Customer Satisfaction: 5		Customer Dissatisfaction: 1
Supporting Material: None		Conflict: None
History: 14/12/24		

Requirement # 3	Requirement Type: FR	Event/Use Case: View Appointments
Description: Admin can view all appointments		
Rationale: Provides admin with an overview of daily activities.		
Originator: Team Member		
Fit Criterion: Admin can see the list of all scheduled appointments.		
Priority: High		
Customer Satisfaction: 5	Customer Dissatisfaction: 2	
Supporting Material: None	Conflict: None	
History: 14/12/24		

Requirement # 4	Requirement Type: FR	Event/Use Case: Search for Car
Description: Admin can search for a car using its unique car number to retrieve service history.		
Rationale: Helps track car services and customer records.		
Originator: Stakeholder		
Fit Criterion: Admin can search and retrieve car records using car number.		
Priority: Medium		
Customer Satisfaction: 4	Customer Dissatisfaction: 3	
Supporting Material: None	Conflict: None	
History: 15/12/24		

Requirement # 5	Requirement Type: FR	Event/Use Case: View Financial Reports
Description: Admin can view reports showing income, expenses, and profit		
Rationale: Provides insight into the financial performance of the business.		
Originator: Team Member		
Fit Criterion: Financial report is generated showing income, expenses, and profit.		
Priority: High		
Customer Satisfaction: 5		Customer Dissatisfaction: 2
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 6	Requirement Type: FR	Event/Use Case: Manage Clients
Description: Admin can manage client records (add, update, delete).		
Rationale: Helps keep customer information up to date.		
Originator: Stakeholder		
Fit Criterion: Client details can be added, updated, or removed.		
Priority: High		
Customer Satisfaction: 5		Customer Dissatisfaction: 2
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 7 Requirement Type: FR Event/Use Case: Send Notifications

Description: Admin can send reminders to clients regarding due services.

Rationale: Ensures clients are reminded about their service schedules.

Originator: Stakeholder

Fit Criterion: Clients receive notifications at the scheduled time.

Priority: High

Customer Satisfaction: 5

Customer Dissatisfaction: 2

Supporting Material: None

Conflict: None

History: 15/12/24

Requirement # 8 Requirement Type: FR Event/Use Case: Schedule Due Services

Description: Admin can view due services and mark them as false to don't show up on due service.

Rationale: Keeps track of services that need to be done, helping with follow-ups.

Originator: Stakeholder

Fit Criterion: Due services are visible, and completed services can be marked.

Priority: High

Customer Satisfaction: 5

Customer Dissatisfaction: 2

Supporting Material: None

Conflict: None

History: 15/12/24

Requirement # 9	Requirement Type: FR	Event/Use Case: Manage Employees
Description: Admin can manage employee records (add, update, delete).		
Rationale: Ensures employee information is up to date.		
Originator: Team Member		
Fit Criterion: Employee records can be added, updated, or removed.		
Priority: Medium		
Customer Satisfaction: 4		Customer Dissatisfaction: 3
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 10	Requirement Type: FR	Event/Use Case: Employee Task Assignment
Description: Employees can be assigned to tasks for car detailing services.		
Rationale: Helps ensure that the right employees are assigned to each job.		
Originator: Stakeholder		
Fit Criterion: Task assignments are accurately made for employees.		
Priority: High		
Customer Satisfaction: 5		Customer Dissatisfaction: 2
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 11	Requirement Type: FR	Event/Use Case: Appointment Completion
Description: Employees mark appointments as completed after finishing the services.		
Rationale: Ensures the system keeps track of completed tasks.		
Originator: Team Member		
Fit Criterion: Appointment status is updated to completed.		
Priority: High		
Customer Satisfaction: 5		Customer Dissatisfaction: 1
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 12	Requirement Type: FR	Event/Use Case: Client Search by Car Number
Description: Admin and Employee can search for a client based on the car number.		
Rationale: Allows quick retrieval of client data linked to their car.		
Originator: Stakeholder		
Fit Criterion: Client records can be found by searching car number.		
Priority: High		
Customer Satisfaction: 5		Customer Dissatisfaction: 2
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 13	Requirement Type: FR	Event/Use Case: Appointment Summary
Description: Upon completion, employees can see a summary of the appointment including client and car details.		
Rationale: Ensures transparency and confirmation of completed tasks.		
Originator: Team Member		
Fit Criterion: Summary is shown for completed appointments.		
Priority: Medium		
Customer Satisfaction: 4		Customer Dissatisfaction: 3
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 14	Requirement Type: FR	Event/Use Case: Add Expense
Description: Admin can add expenses for the business operations.		
Rationale: Tracks business expenditures to maintain financial clarity.		
Originator: Stakeholder		
Fit Criterion: Expense records can be added to the system.		
Priority: Medium		
Customer Satisfaction: 4		Customer Dissatisfaction: 3
Supporting Material: None		Conflict: None
History: 15/12/24		

2.9.2 Non-Functional Requirement

Requirement # 1	Requirement Type: Non-FR	Event/Use Case: System Response
Description: The system must support up to 25 concurrent users without degradation in performance.		
Rationale: Ensures smooth operation during peak hours.		
Originator: Stakeholder		
Fit Criterion: System handles 25 users simultaneously under normal load.		
Priority: High		
Customer Satisfaction: 8		Customer Dissatisfaction: 9
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 2	Requirement Type: Non-FR	Event/Use Case: System Uptime
Description: The system must maintain an uptime of 95%.		
Rationale: Ensures continuous service availability.		
Originator: Team Member		
Fit Criterion: System is available 95% of the time in a month.		
Priority: High		
Customer Satisfaction: 8		Customer Dissatisfaction: 8
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 3	Requirement Type: Non-FR	Event/Use Case: User Interaction
Description: The system should respond to any user query or action within 5 seconds under normal load.		
Rationale: Provides a fast and responsive user experience. Ensures continuous service availability.		
Originator: Stakeholder		
Fit Criterion: Response time for actions is under 5 seconds.		
Priority: High		
Customer Satisfaction: 9		Customer Dissatisfaction: 9
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 4	Requirement Type: Non-FR	Event/Use Case: User Interface
Description: The interface should be easy to navigate and require minimal training for new employees.		
Rationale: Ensures user adoption and reduces training time.		
Originator: Stakeholder		
Fit Criterion: 90% of users can use the system without extensive training.		
Priority: Medium		
Customer Satisfaction: 8		Customer Dissatisfaction: 7
Supporting Material: None		Conflict: None
History: 15/12/24		

Requirement # 5	Requirement Type: Non-FR	Event/Use Case: System Requirements
Description: The system must run on Windows 10 and above, with 4 GB RAM and 2 GB storage.		
Rationale: Ensures the software works on existing hardware infrastructure.		
Originator: Team Member		
Fit Criterion: System is operational on Windows 10+ with defined hardware.		
Priority: Medium		
Customer Satisfaction: 7		Customer Dissatisfaction: 8
Supporting Material: None		Conflict: None
History: 15/12/24		

Chapter 3 Software Process Model

3.1. Software Process Models

A software process model defines the framework for planning, structuring, and controlling the development of an information system. Software process models serve as a guide for software development, ensuring that the development process is systematic, predictable, and efficient.

3.1.1 Common Software Process Models

Various software process models are used globally; each suited to different types of projects. Below are the most prominent ones:

1. Waterfall Model
2. V-Model
3. Incremental Model
4. Spiral Model
5. Agile Model
6. DevOps Model
7. Prototype Model

3.2 Proposed Model with diagram

For the **Pristine Shine Desktop Application**, we have chosen to implement the **Agile Model** using the **Scrum Framework**. The Scrum framework focuses on short iterations, known as sprints, where each sprint is a mini-project that produces a working version of the software.

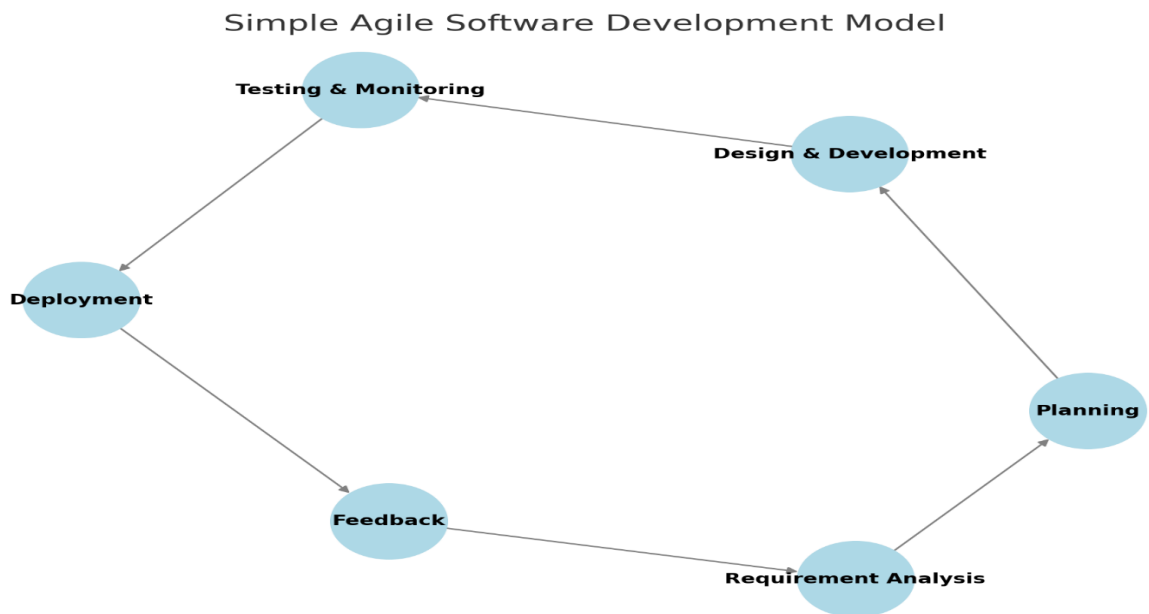


Figure 4 Agile Model Diagram

While all the above models have their merits, the Agile Model was selected for the Pristine Shine Desktop Application due to its ability to adapt quickly to changing requirements, its focus on customer collaboration, and its suitability for iterative development and feedback. This model is particularly beneficial when dealing with complex, evolving requirements and when frequent releases of working software are essential.

3.2.1 Reason

The decision to use the Agile Model, specifically Scrum, for the Pristine Shine Desktop Application, is based on several factors:

1. **Dynamic Requirements:** The business requirements for **Pristine Shine** are expected to evolve as the project progresses. Agile allows for changes in the scope of work and adapts to new priorities as they arise.
2. **Customer Involvement:** With Agile, we can involve stakeholders throughout the process, ensuring their needs are continuously met. This aligns with the goal of developing a customer-centric application.
3. **Collaboration:** Agile promotes constant communication among team members, stakeholders, and users. This helps identify and resolve issues early.
4. **Quick Delivery:** The Scrum approach allows us to deliver functional versions of the system at the end of every sprint, ensuring that the system is continuously evolving and delivering value.
5. **Early Risk Mitigation:** As features are developed and tested incrementally, potential risks are identified and resolved early in the process, reducing the chances of project failure.

3.3 Implementation of the Agile Model

The Agile Model is implemented through various phases, which ensure that the project progresses in small, manageable increments, allowing for continuous feedback and improvement.

3.3.1 Requirement Analysis

Requirement analysis is the first step in the Agile process where the development team collaborates with stakeholders to understand the needs and expectations of the end-users. This phase focuses on gathering and defining the project's functional and non-functional requirements. In Agile, these requirements are typically broken down into smaller user stories, which serve as the foundation for each sprint. The requirement analysis is iterative, and as more information becomes available during development

Objective:

To gather and define both functional and non-functional requirements from stakeholders.

Activities:

- Meetings with stakeholders to understand business needs.

- Documentation of functional and non-functional requirements.
- Initial high-level design ideas for the system's core functionality.

Deliverables:

- **Functional Requirements Document:** Contains all features and functionalities that the system should have.
- **Non-Functional Requirements Document:** Describes performance, compatibility, and other system constraints.
- **Use Case Diagrams:** Visual representations of system interactions.

3.3.2 Planning of Sprint

In Agile, work is divided into manageable chunks called sprints. The planning of a sprint involves selecting user stories from the product backlog (which includes all prioritized user stories) and estimating the effort required to complete them. During this stage, the team discusses the scope, objectives, and deliverables for each sprint. The goal is to ensure that the team has a clear understanding of what needs to be done and the timeline for the sprint.

Objective

Plan tasks for the sprint, prioritize work, and allocate responsibilities.

Activities:

- Define a sprint goal (usually 2-4 weeks).
- Break down features into smaller tasks for implementation.
- Identify critical milestones and deadlines.

Deliverables:

- **Sprint Backlog:** A list of tasks and features for the current sprint.
- **Task Allocation:** Assign developers and testers to the tasks.

3.3.3 Collaborative Design and Development

Agile emphasizes collaboration among cross-functional teams, including developers, designers, business analysts, and testers. During the design and development phase, team members work together in an iterative and incremental manner. The focus is on creating a minimum viable product (MVP) that can be quickly tested and iterated upon.

Objective:

Design and develop the application collaboratively.

Activities:

- Collaborative design of system components (UI, database, etc.).
- Code reviews and pair programming to ensure quality.

- Iterative development of core features, such as user management, appointment booking, and reporting.

Deliverables:

- **Working Prototypes:** Regular prototypes of the application at the end of each sprint.
- **System Design Documentation:** Updated design documents after each sprint.

3.3.4 Implementation

Implementation in Agile refers to the actual development work, where code is written, features are built, and the system starts to take shape. As the team progresses through the sprints, they continuously integrate the new code with the existing system to ensure the product is functional and complete. Agile focuses on delivering usable and functioning software at the end of each sprint, allowing for immediate feedback from stakeholders.

Objective:

Integrate and deploy features into the working system.

Activities:

- Full system integration after each sprint.
- Deployment of completed features for internal testing.

Deliverables:

- **Deployed Features:** Incremental releases of completed features.

3.3.5 Testing and Monitoring

Objective:

Ensure software quality through continuous testing and monitoring.

Activities:

- Conduct unit testing, integration testing, and system testing for each feature.
- Monitor the system post-deployment to ensure smooth operation.

Deliverables:

- **Test Reports:** Results from each phase of testing.
- **Bug and Issue Logs:** A record of any bugs or issues encountered during testing.

Chapter 4: PROPOSED SYSTEM

4.1 Need for the Proposed System

The lack of digitalization in many car detailing studios like Pristine Shine leads to inefficient manual processes such as booking services, managing client records, tracking financials, and providing service reminders. This often results in missed appointments, disorganized client data, poor follow-ups, and lack of insight into overall business performance.

Furthermore, managing such operations manually increases the risk of data loss, human error, and time consumption. The growing need for automation, customer retention, and financial tracking in the car detailing sector drives the need for a centralized desktop-based software solution.

4.2 Features of Proposed System

1. Appointment Booking & Scheduling

Allows employees and admin to book services for customers, assign employees, apply discounts, and mark payment methods. Status is set to “Pending” or “Completed”.

2. Client and Car Record Management

Stores detailed client profiles including contact info, CNIC, and multiple cars per client. Also supports multiple owners for a single vehicle.

3. Search Functionality

Advanced car search using CNIC or car number retrieves:

- Total services done
- Last service date
- Car details (model, year, make, color)
- Complete appointment/service history

4. Due Service Reminder System

The system filters cars due for service (e.g., after 30 days) and sends SMS reminders to clients via the integrated Twilio API using a single-click notification feature.

5. Services Module

Manage and update service packages such as Basic Wash, Interior Detailing, Ceramic Coating, etc. Admin can add/edit service details and pricing.

6. Employee Management

Admin can add or update employee records with details such as:

- Name
- Email
- CNIC
- Role
- Salary
- Address

7. Financial Reporting

Generates income and expense reports, enabling business performance analysis over time. Filters allow report generation based on date ranges.

8. User and Session Management

Displays currently logged-in users, session control, and logout functionality for both admin and employees.

9. Dashboard

The homepage shows a quick summary:

- Total clients
- Total services performed
- Daily/Monthly income
- Upcoming appointments
- Graphical visualizations

4.3 Business Context

4.3.1 Advantages in Business

Business Benefit	Description
Improved Efficiency	Streamlines service booking, client management, and billing.
Better Client Retention	Automatic follow-ups improve customer loyalty.
Organized Data Handling	Digital records reduce errors and loss.
Time-Saving	Quick search and automation save hours of manual work.
Financial Transparency	Real-time tracking of income, expenses, and profits.
Competitive Advantage	Modern, tech-enabled operations increase business credibility.

Table 8 Advantages in Business

4.3.2 Stakeholder's Profiles

Stakeholder	Major Value	Attitudes
Owner/Admin	Full control, profit insights	Sees system as professional and scalable
Employees	Easy booking & record keeping	Trusts system for reducing workload
Clients	Personalized, timely service reminders	Appreciate professional management
Developer	Maintains and scales system	Believes in automation and innovation

Table 9 Stakeholder's Profiles

4.3.3 Project Priorities

Dimension	Driver	Constraint
Schedule	Ready before graduation	All modules to be functional
Features	Service scheduling, reminders	All key modules in version 1.0
Quality	Reliable and responsive	Accuracy in financial calculations
Cost	Must be free to deploy	Open-source dependencies

Table 10 Project Priorities

4.4 Operating Environment

Component	Specification
Platform	Windows Desktop Application (WinForms)
Backend Database	MySQL 9.1 (runs as Windows service)
Dependencies	.NET Framework 4.8, Python 3.11.5
AI Integration	Python script with OpenAI API
SMS Gateway	MoceanAPI for automated client reminders
Installer	Inno Setup (Offline full installer)

Table 11 Operating Environment

- All installations including MySQL, Python, and the app are handled via a bundled
- System runs offline with no user interaction required for setup.
- Database auto-starts with Windows; no need for manual service activation.
- Minimal system requirements: 4GB RAM, 1GB free disk, .NET Runtime pre-installed.

CHAPTER 5: SYSTEM DESIGN

5.1 Brief Description

System design is a critical phase in the software development lifecycle where the structure and interaction of system components are defined. In this chapter, we present the architectural blueprint of the *Pristine Shine* desktop application, covering its database design, user interface planning, and functional interactions. It translates the gathered requirements into technical models that guide implementation and ensure maintainability.

5.2 UI/UX Design

The user interface was designed with simplicity, responsiveness, and ease of use in mind. The application uses Guna.UI2.WinForms for a modern, animated UI experience. Key principles followed in UI/UX design include:

- **Consistency** in layout and navigation
- **Clear Feedback** through labels and toast messages
- **Error Prevention** via validation on forms
- **Mobile-inspired Animations** using Guna transitions

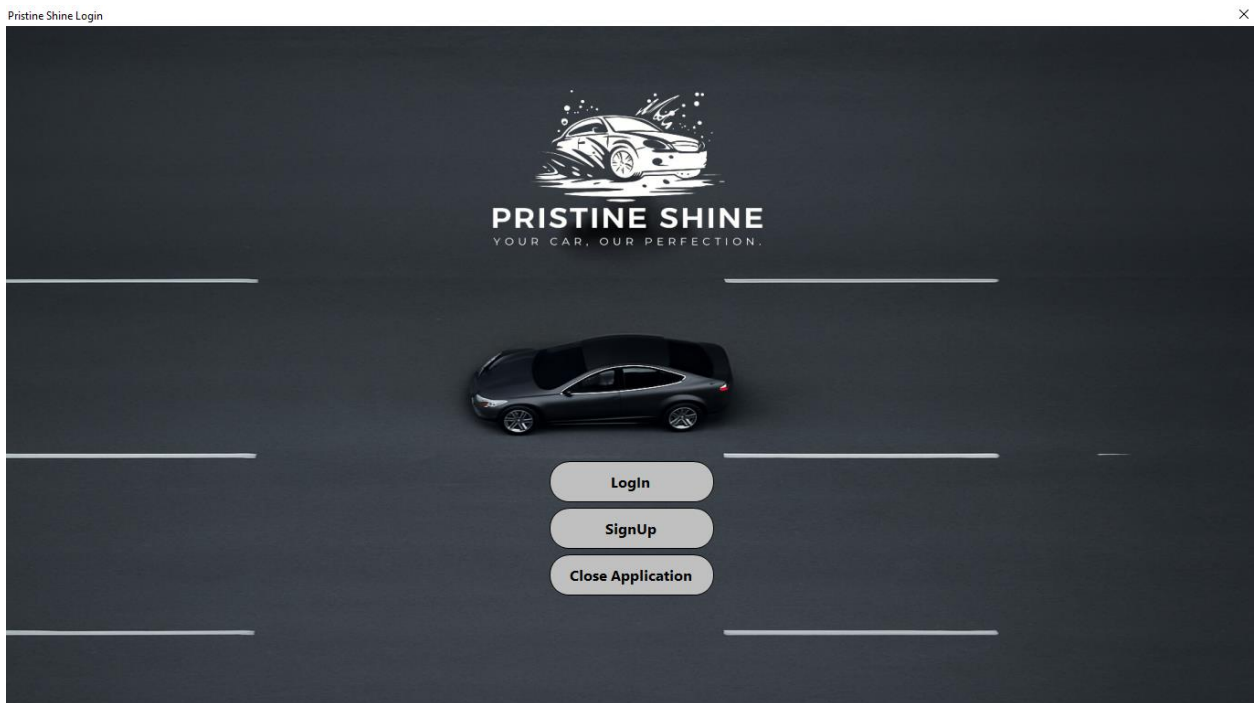


Figure 5 Login Page Screen Shot

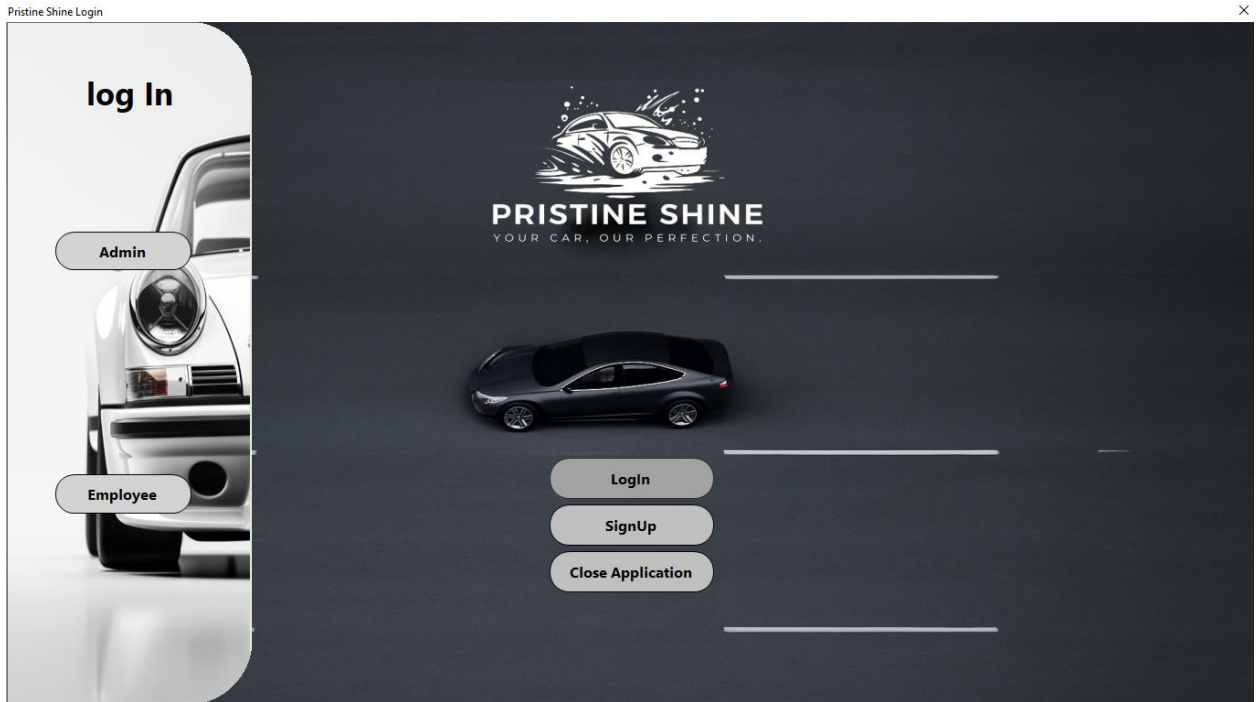


Figure 6 Login Page Screen Shot 2

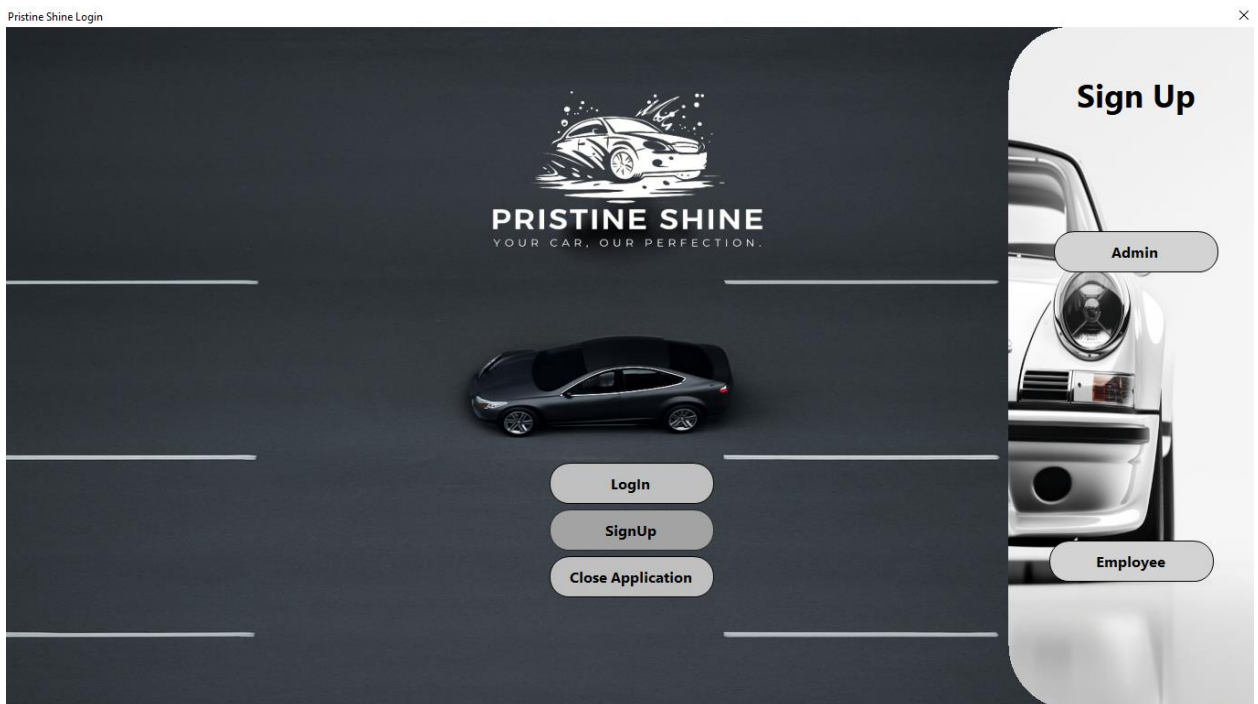


Figure 7 Login Page Screen Shot 3

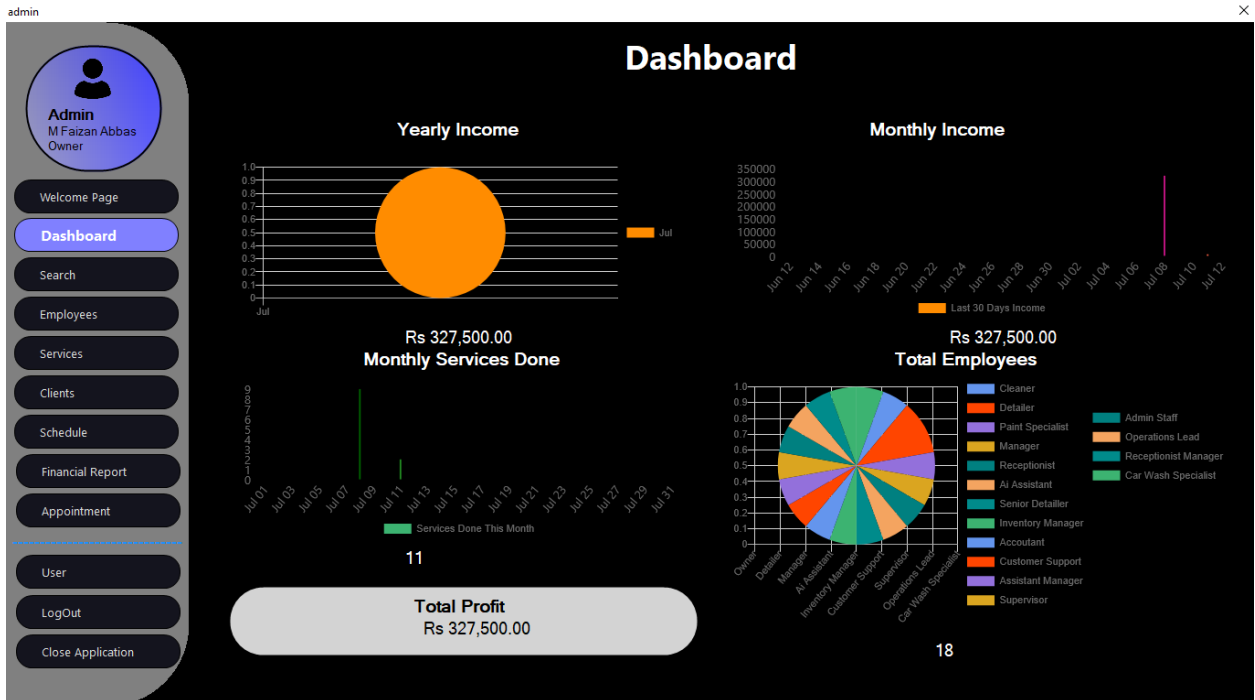


Figure 8 Admin Dashboard Screen Shot

The search screen displays the following information:

- User Profile:** Admin, M Faizan Abbas, Owner.
- Navigation Menu:** Welcome Page, Dashboard, Search, Employees, Services, Clients, Schedule, Financial Report, Appointment, User, LogOut, Close Application.
- Search Bar:** Enter Car Number or Cnic, Search, Clear.
- Results:**
 - Services Done: N/A
 - Last Service Date: N/A
- CAR DATA:** A table with columns for Car Number, Cnic, and Car Name.
- APPOINTMENT DATA:** A table with columns for Appointment Date, Appointment Time, and Appointment Status.

Figure 9 Admin Search Screen Shot

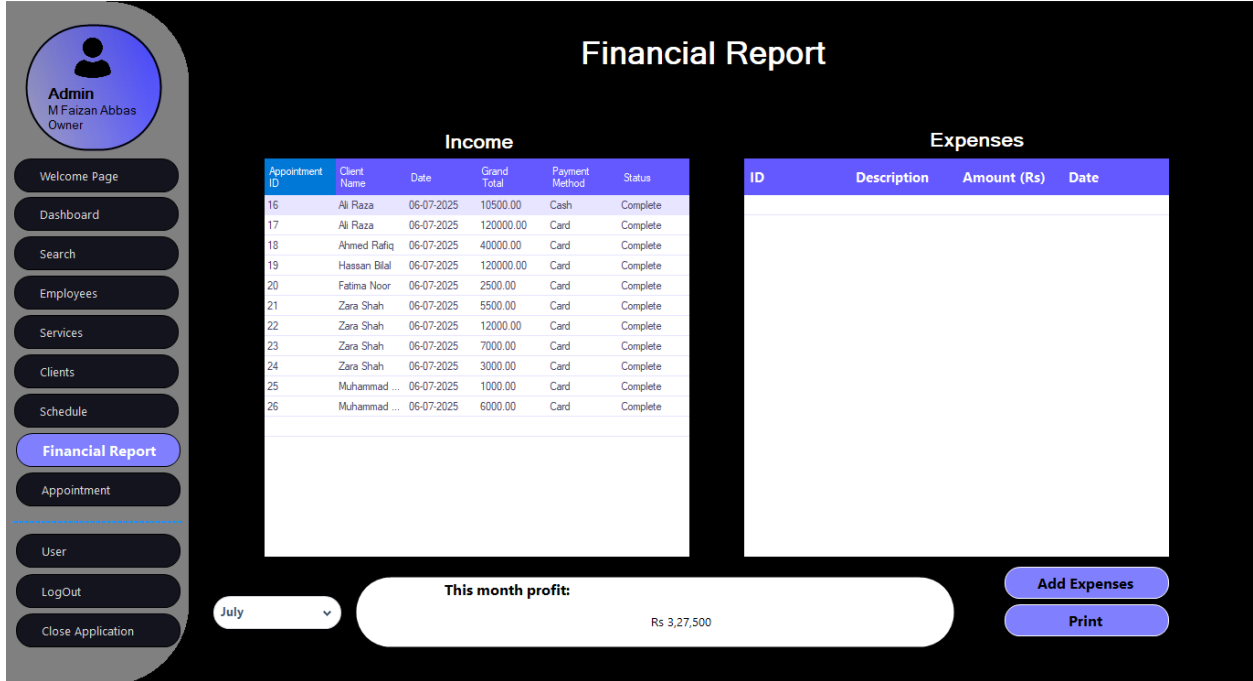


Figure 10 Admin Financial Report Screen Shot

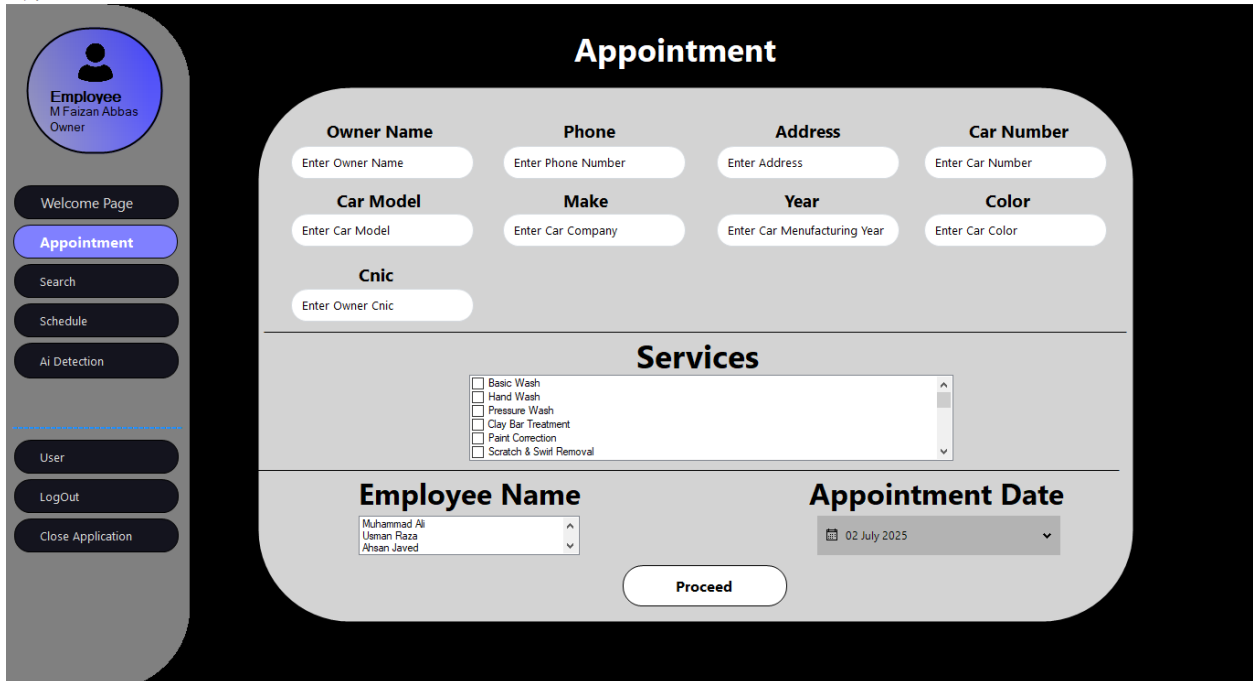


Figure 11 Employee Appointment Page Screen Shot

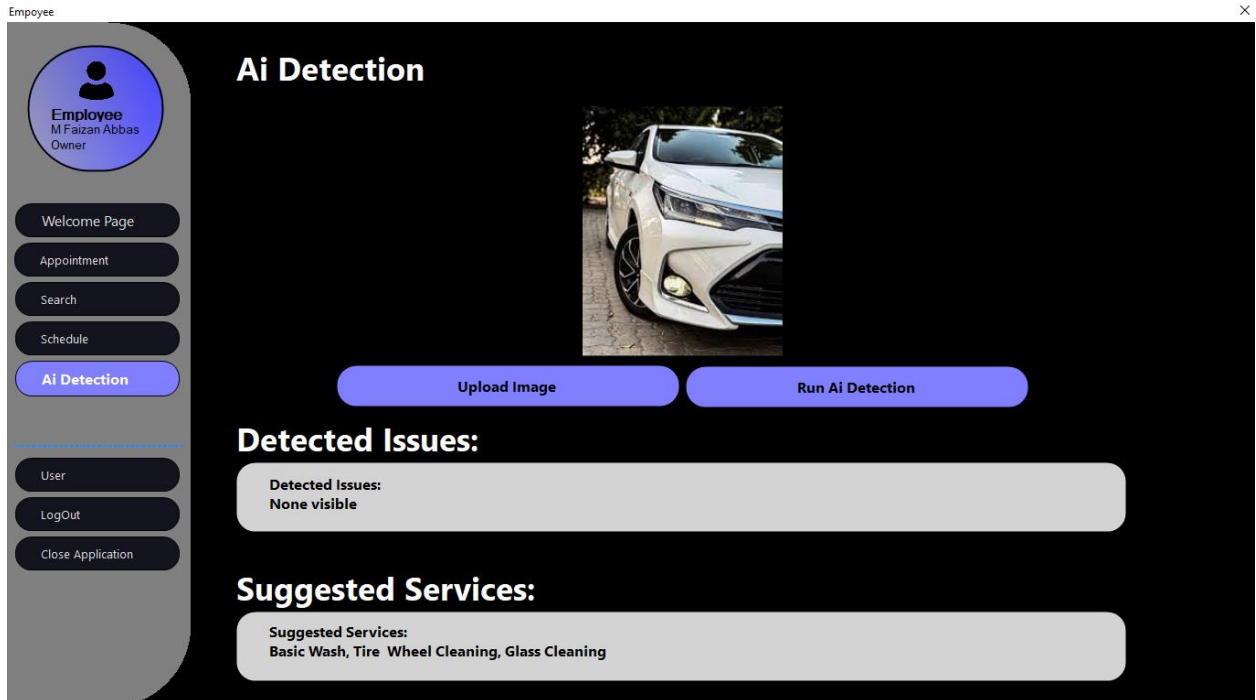


Figure 12 Employee Ai Detection Page Screen Shot

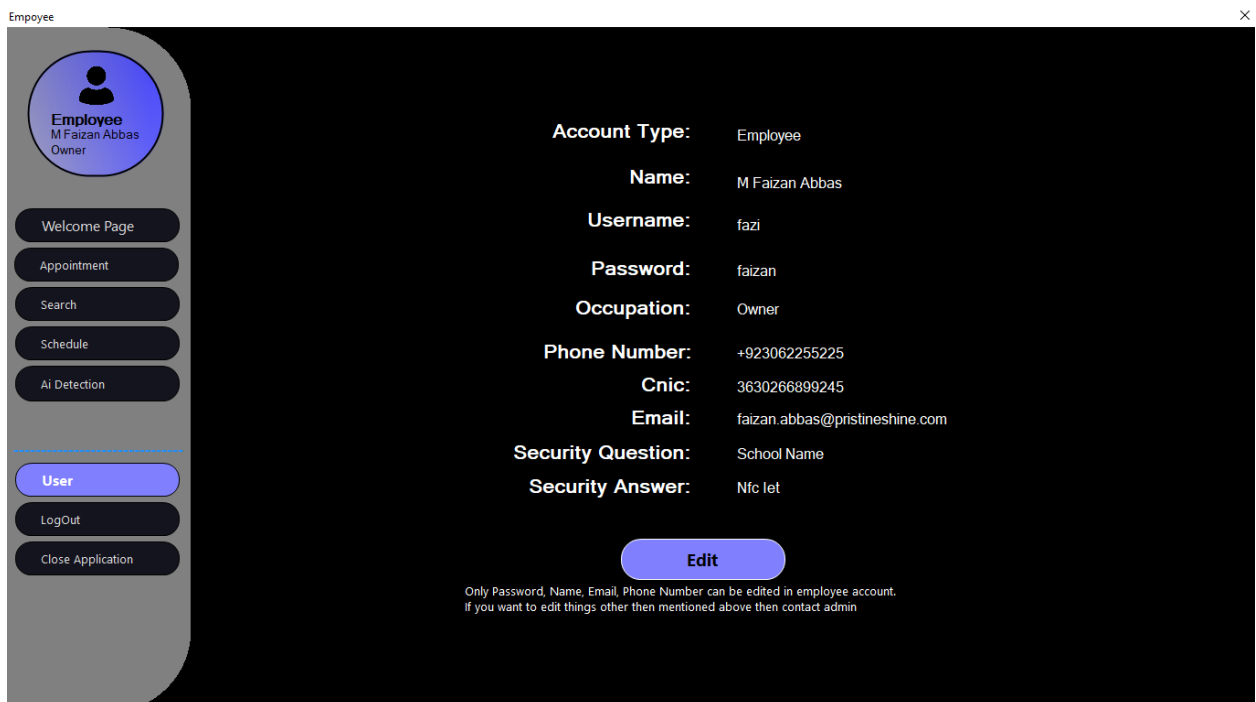


Figure 13 Employee User Page Screen Shot

5.3 ER Diagram

The Entity Relationship (ER) Diagram represents the database schema of *Pristine Shine*. It models all entities, attributes, and their relationships, including:

- Client → owns multiple Cars
- Appointment → links Client and Employee
- Service → assigned to Appointments
- Damage Report → linked to Appointments
- Notification, User, and Financial Report entities

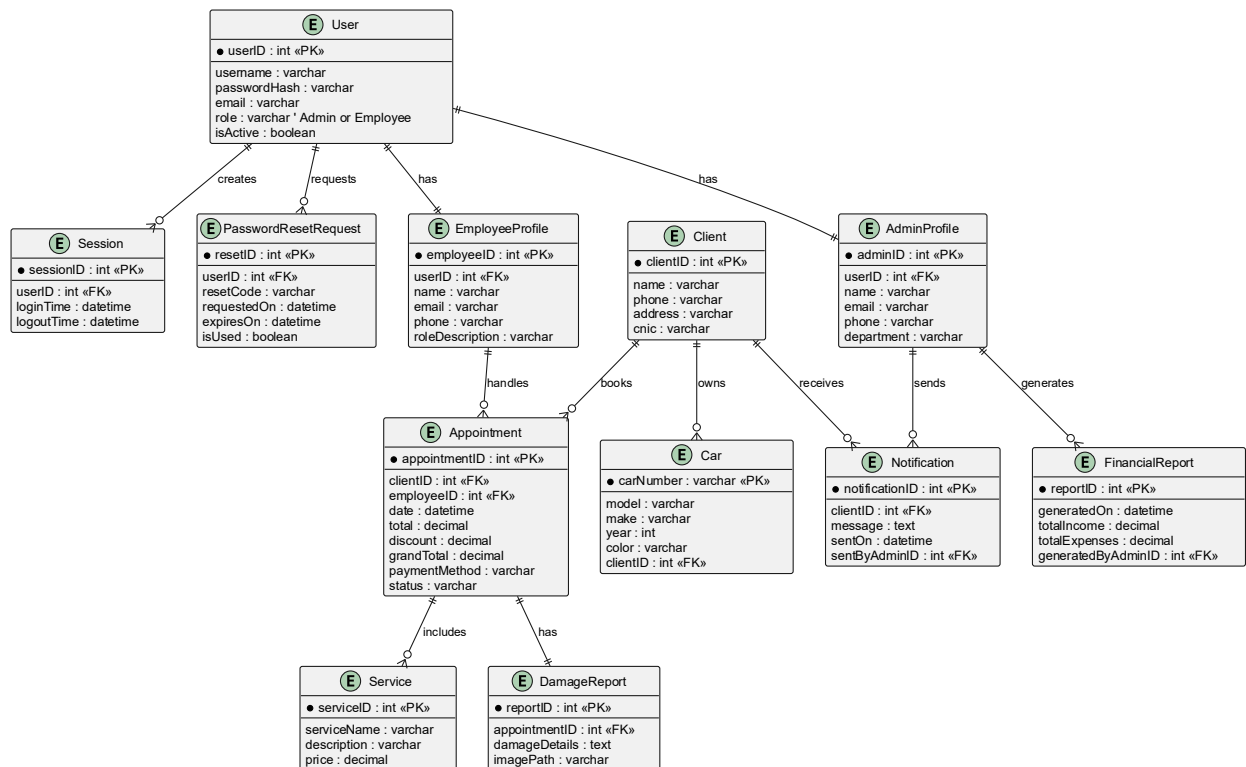


Figure 14 ER Diagram

5.4 Data Flow Diagrams (DFD)

Level 0 - Context Diagram

Shows the system as a single process with external entities like Admin, Employee, and Mocean SMS API interacting with it.

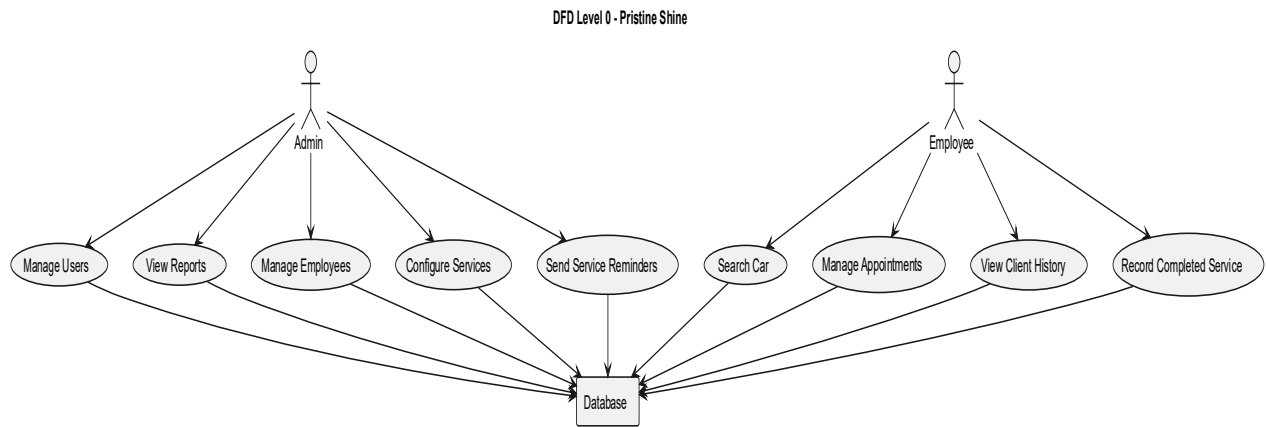


Figure 15 DFD Level-0

Level 1 - System Modules

Breaks the system into main modules:

- Authentication
- Appointment Management
- Service & User Management
- Damage Detection
- Notification Handling
- Financial Reports

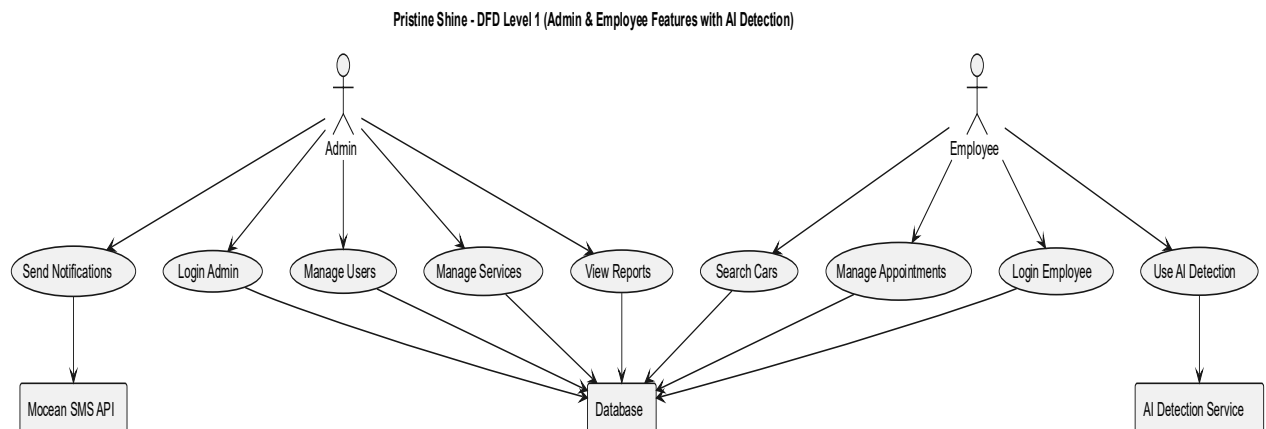


Figure 16 DFD Level-1

Level 2 - Detailed Processes

Level 2 drills down into each module such as:

- Booking Appointments (validation, insertion)
- AI Damage Detection (image processing)
- Sending Reminders (Mocean API)
- Generating Reports (Charts, Income logging)

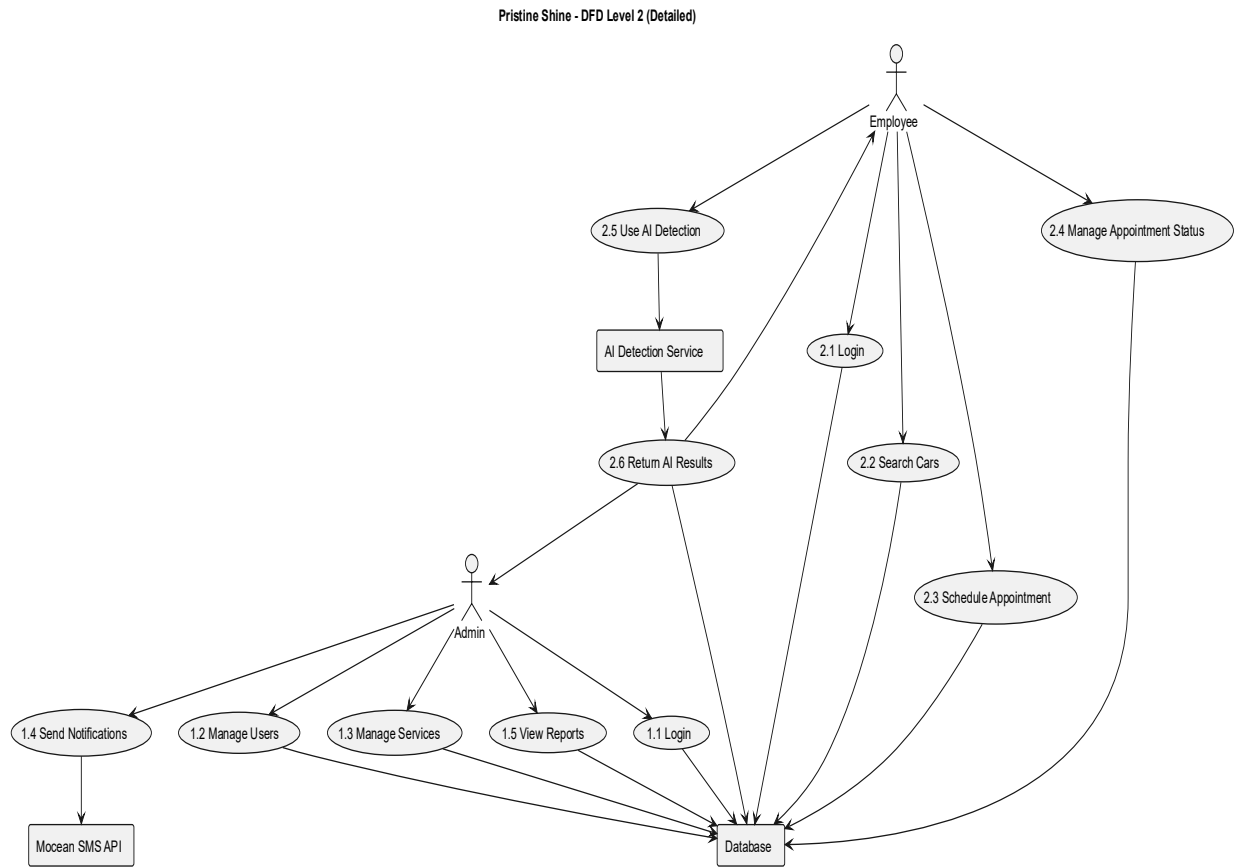


Figure 17 DFD Level-2

5.5 Activity Diagram

Activity diagrams model the flow of user actions. Separate activity flows were designed for:

- Admin Functions (User Management, Report Viewing)
- Employee Functions (Search, Book, AI Detect)
- Login & Session Management
- Forgot Password Flow

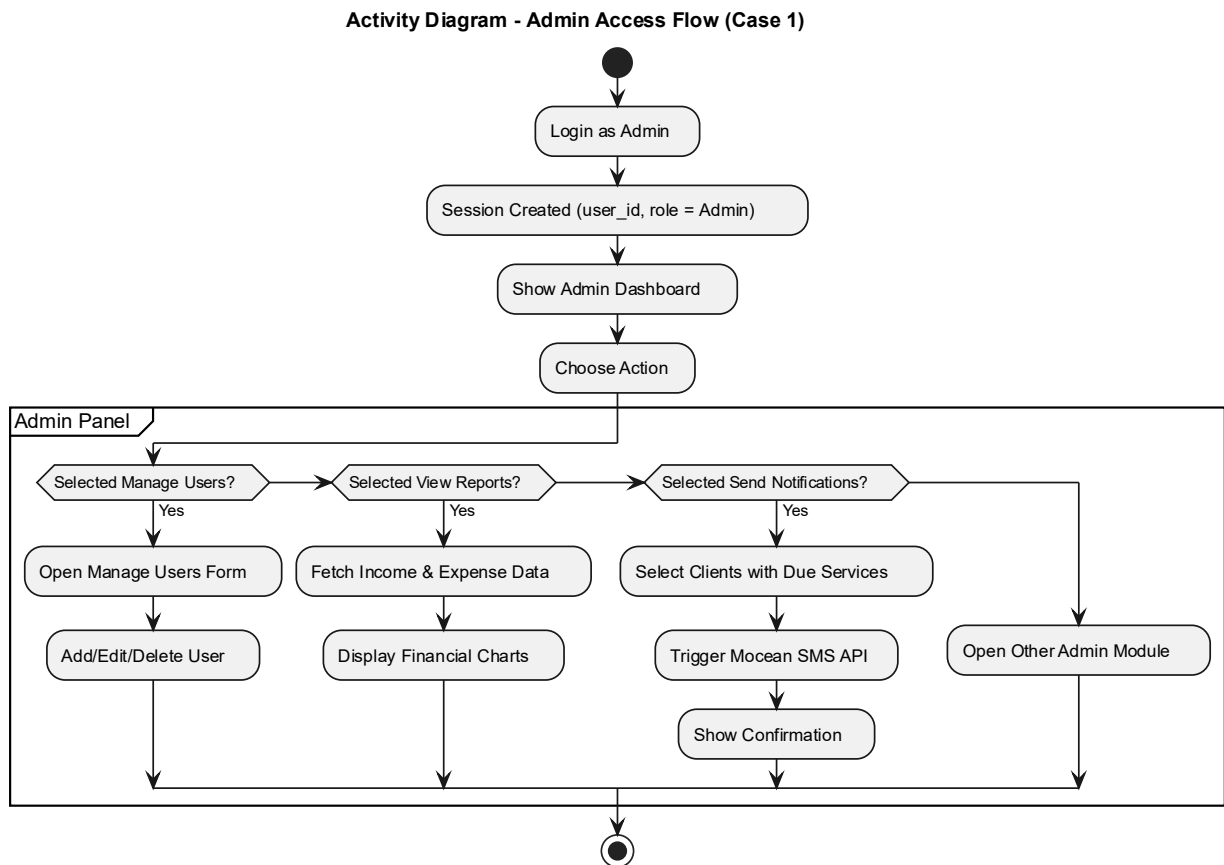


Figure 18 Activity Diagram Case 1

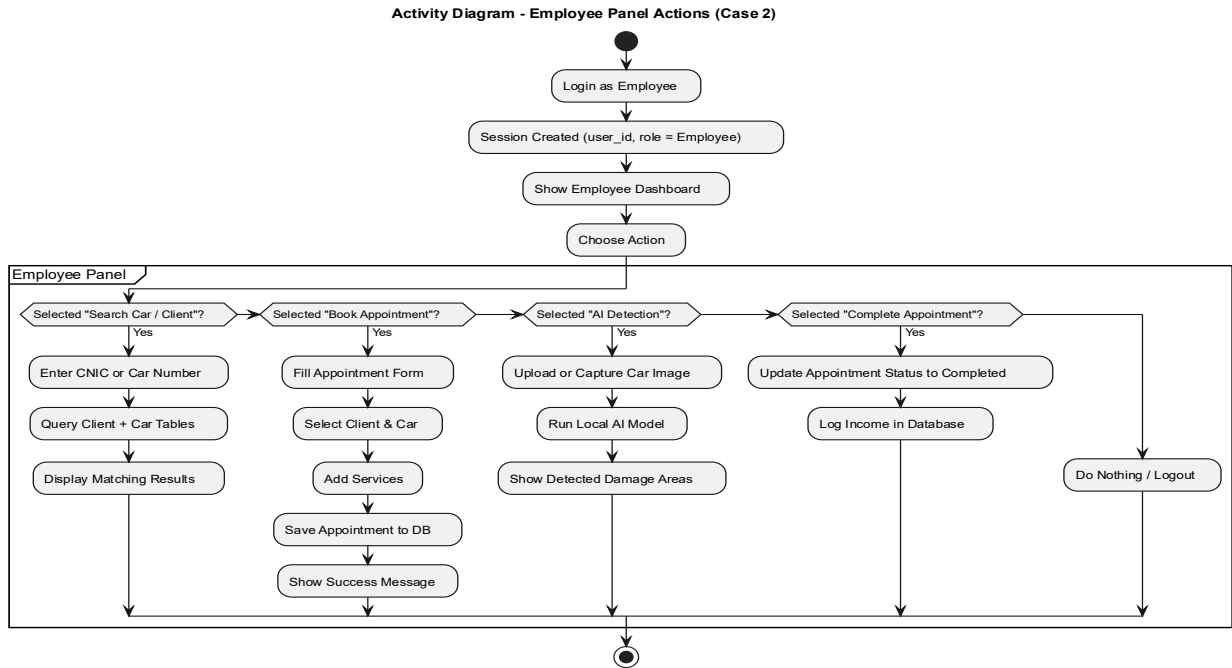


Figure 19 Activity Diagram Case 2

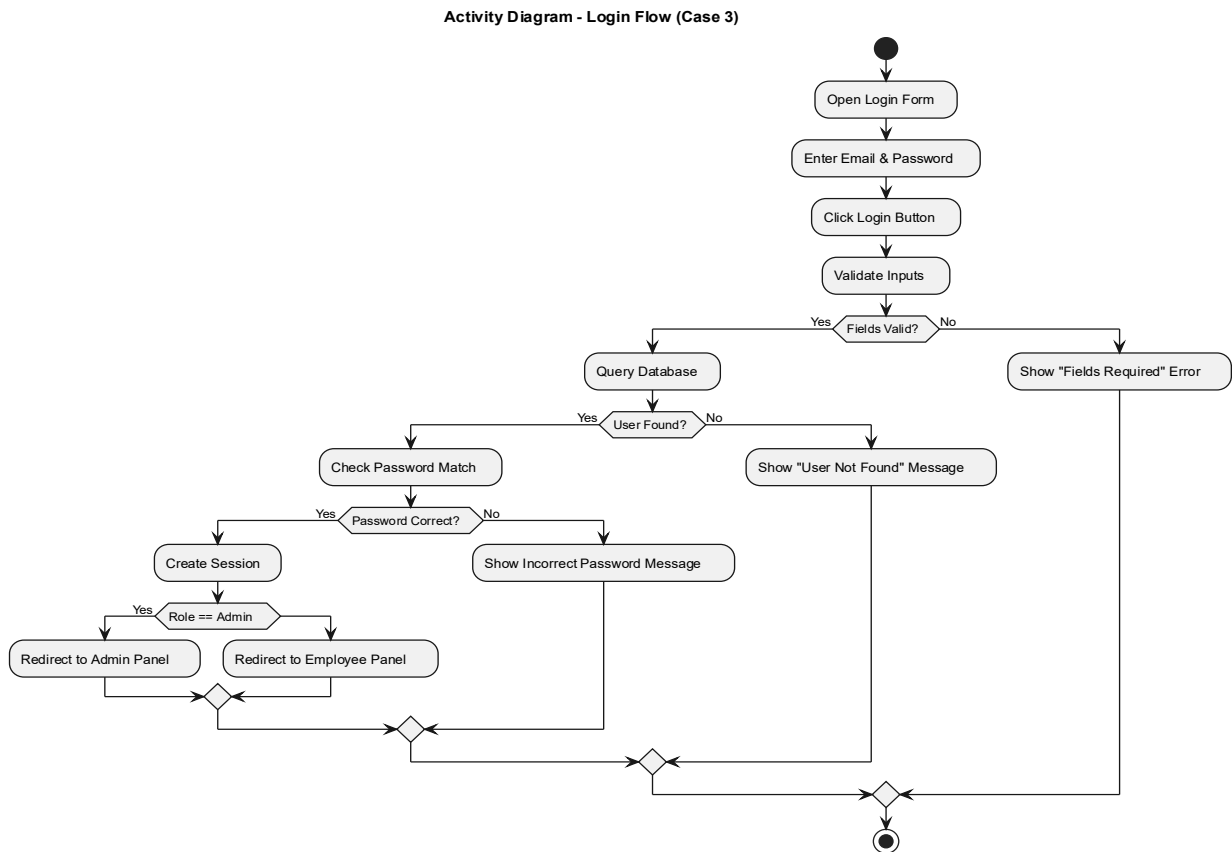


Figure 20 Activity Diagram Case 3

Activity Diagram - Forgot Password (Case 4)

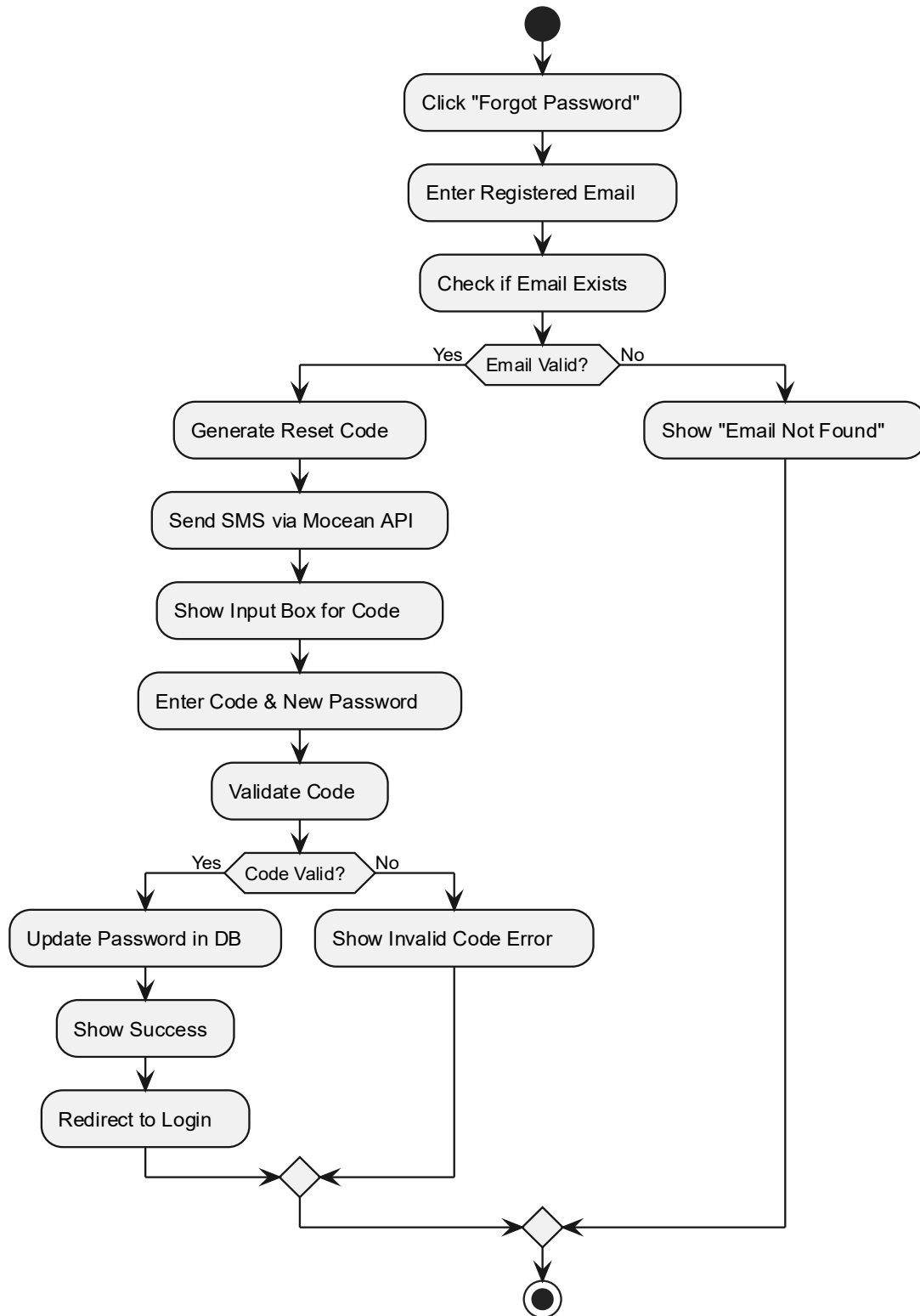


Figure 21 Activity Diagram Case 4

5.6 Class Diagram

The class diagram illustrates system structure through classes and their relationships. Major classes include:

- User, Admin, Employee, Client, Car, Service
- Appointment, Damage Report, Notification
- Relationships such as composition, association, and generalization are applied

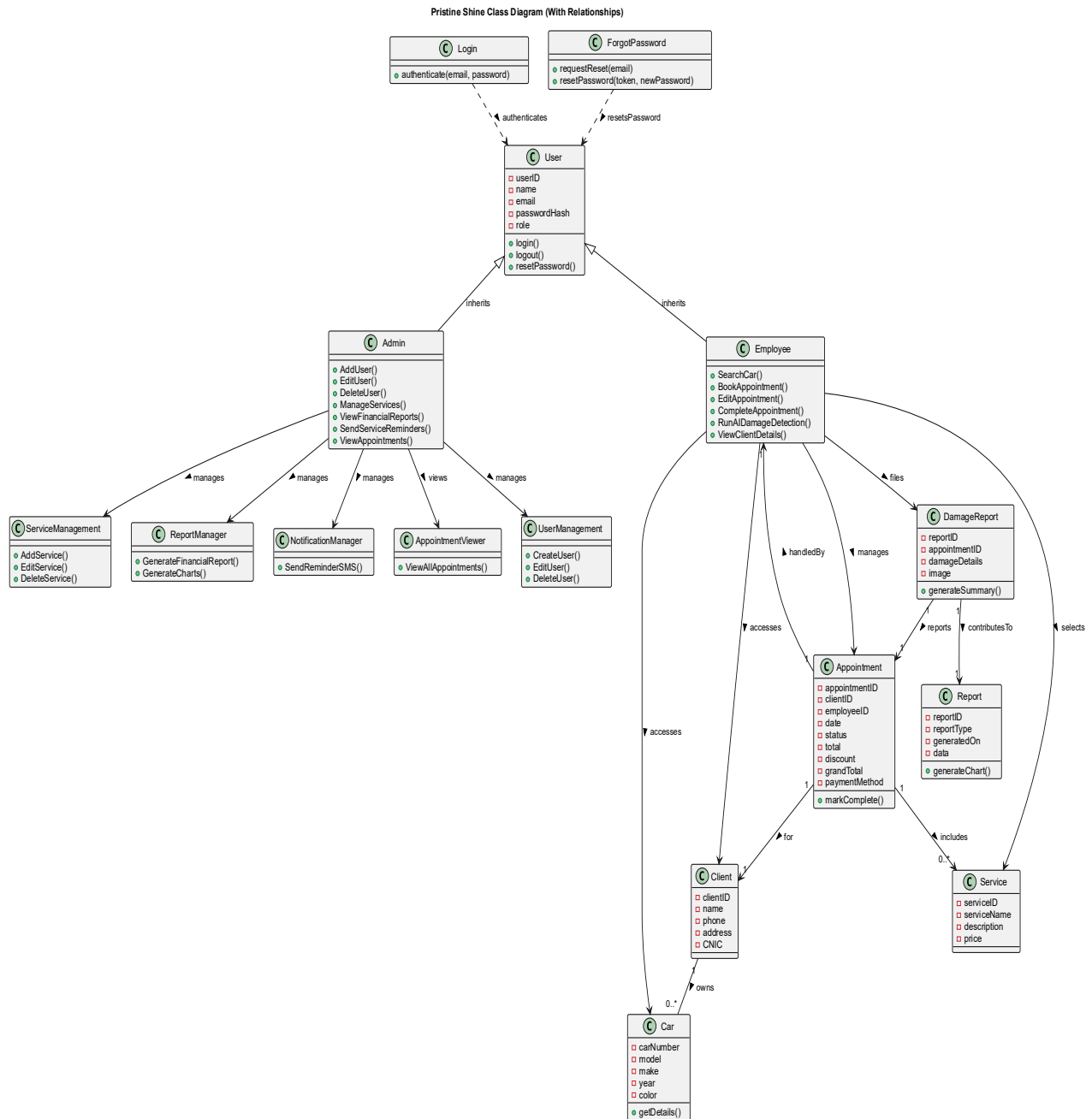


Figure 22 Class Diagram

5.7 Sequence Diagram

Sequence diagrams show the time-based interactions between objects. Designed for:

- Login & Credential Validation
- Appointment Booking Process
- AI Detection Flow
- Admin Sending Notifications

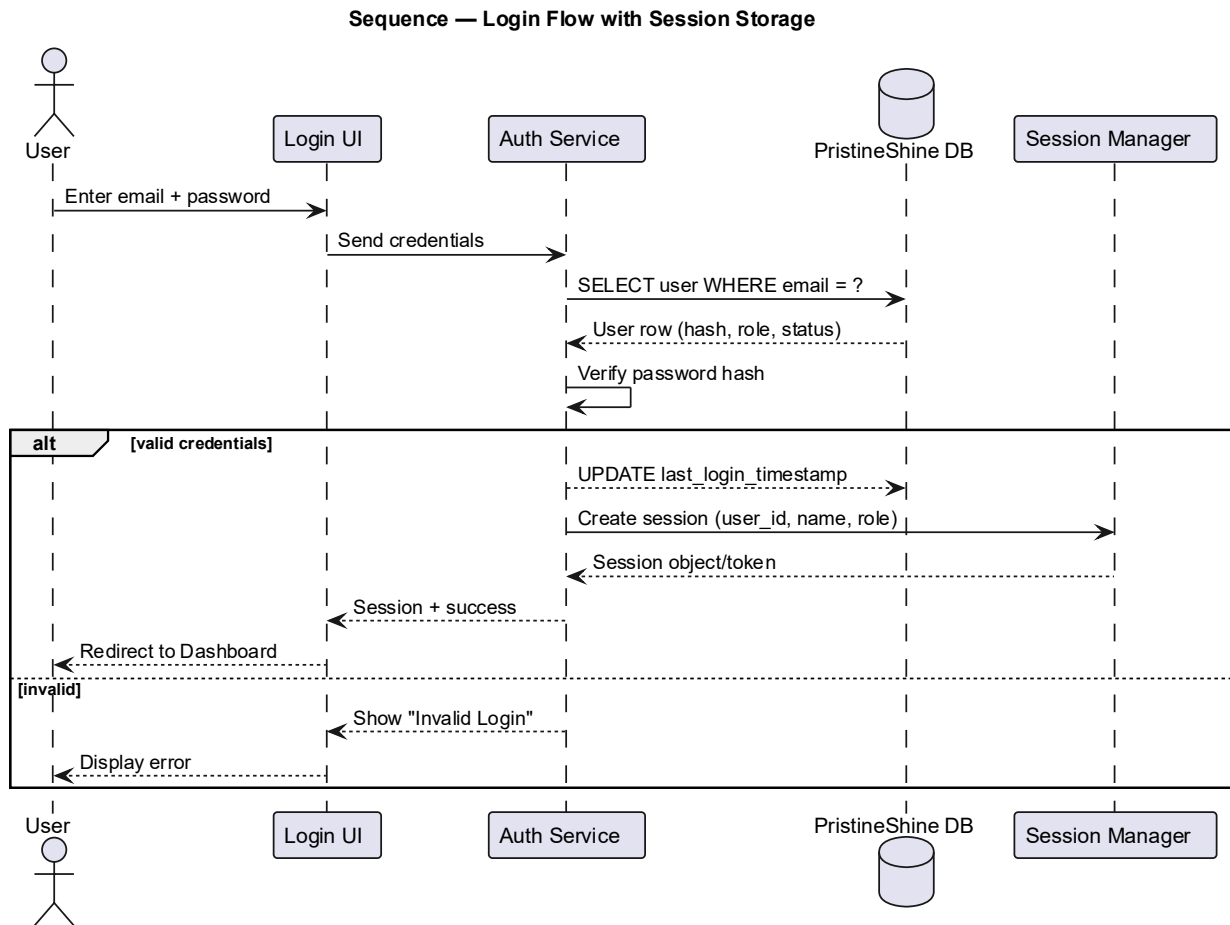


Figure 23 Sequence Diagram - Login

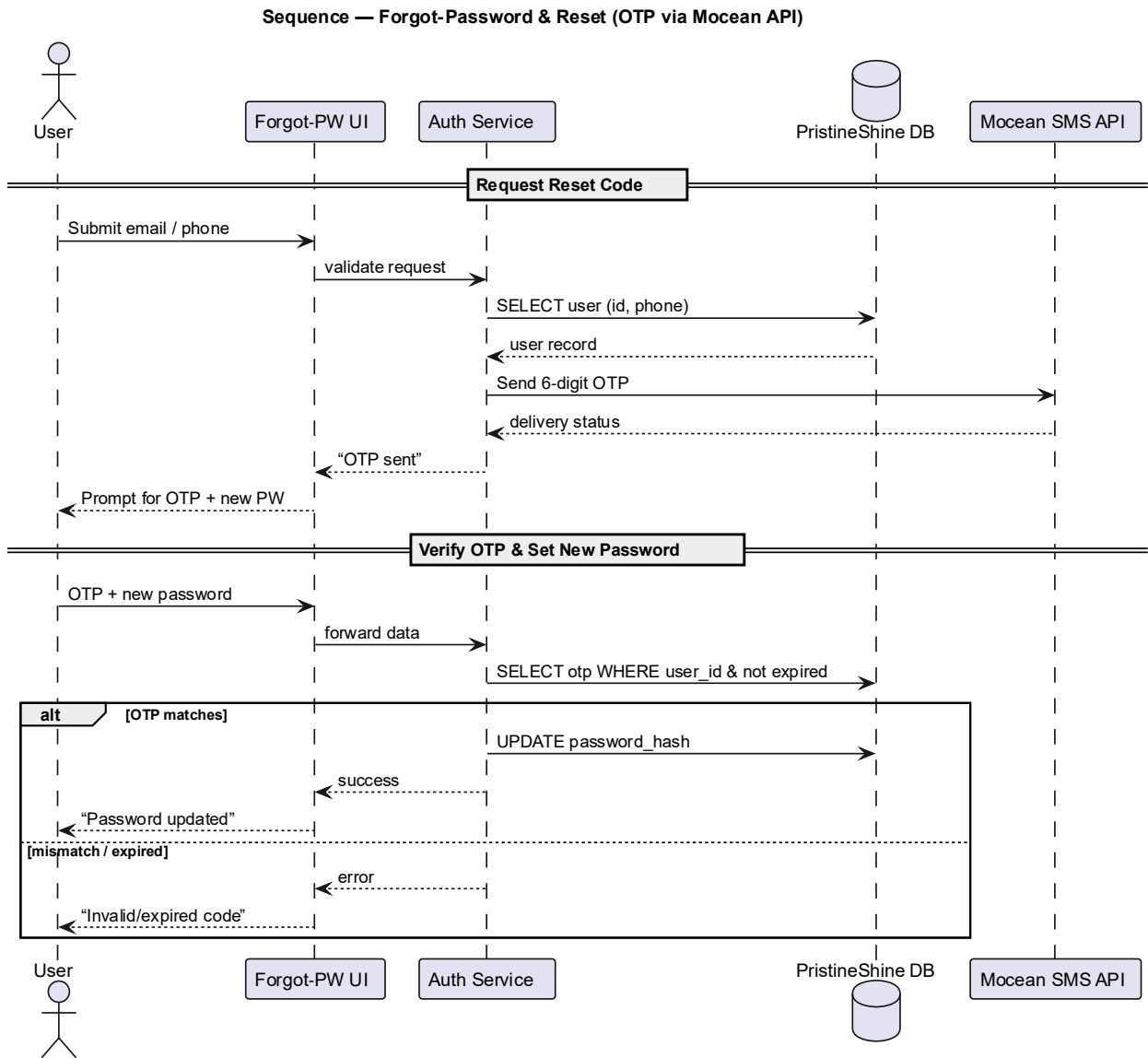


Figure 24 Sequence Diagram -Forgot Password

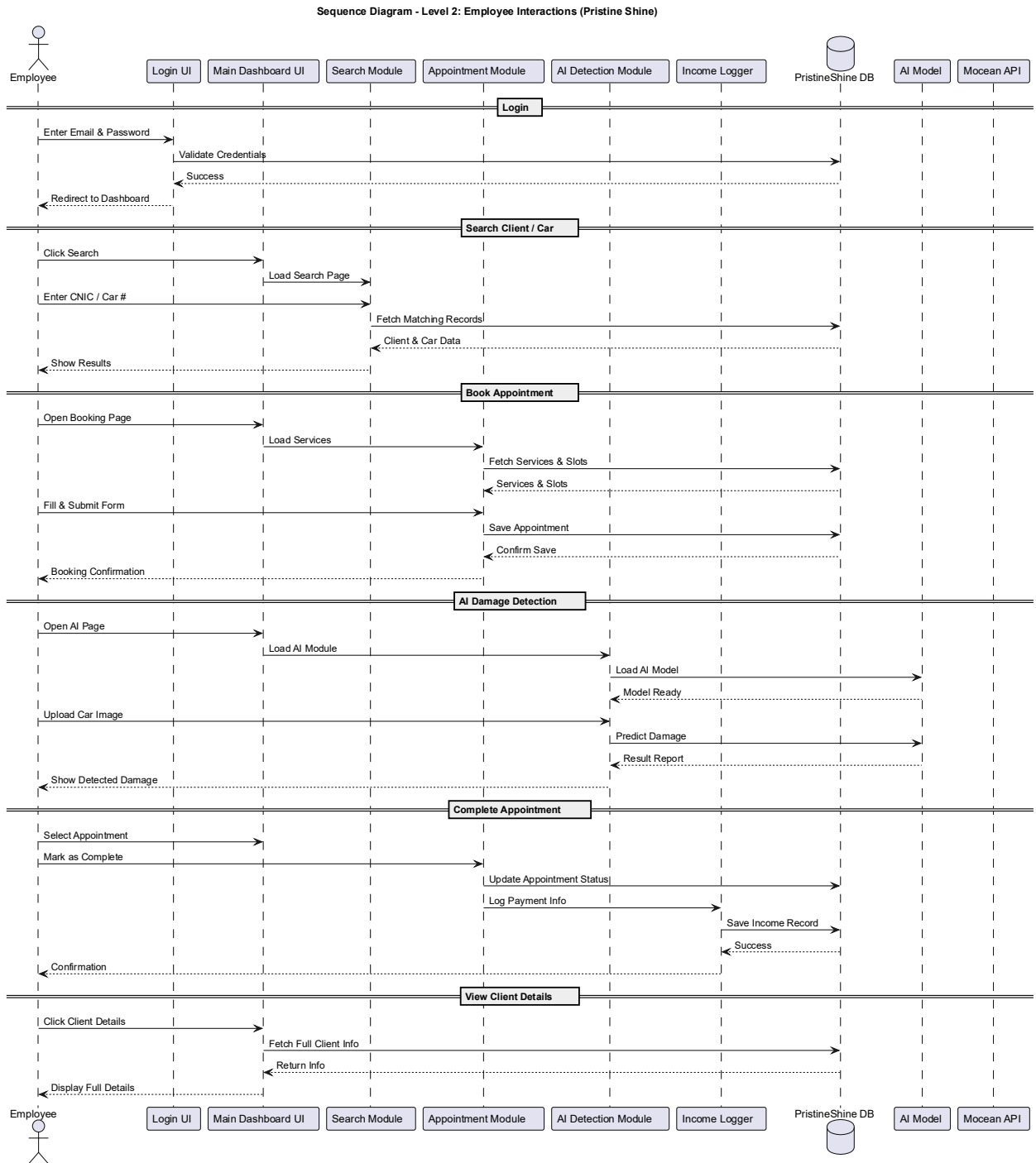


Figure 25 Sequence Diagram -Employee Interactions

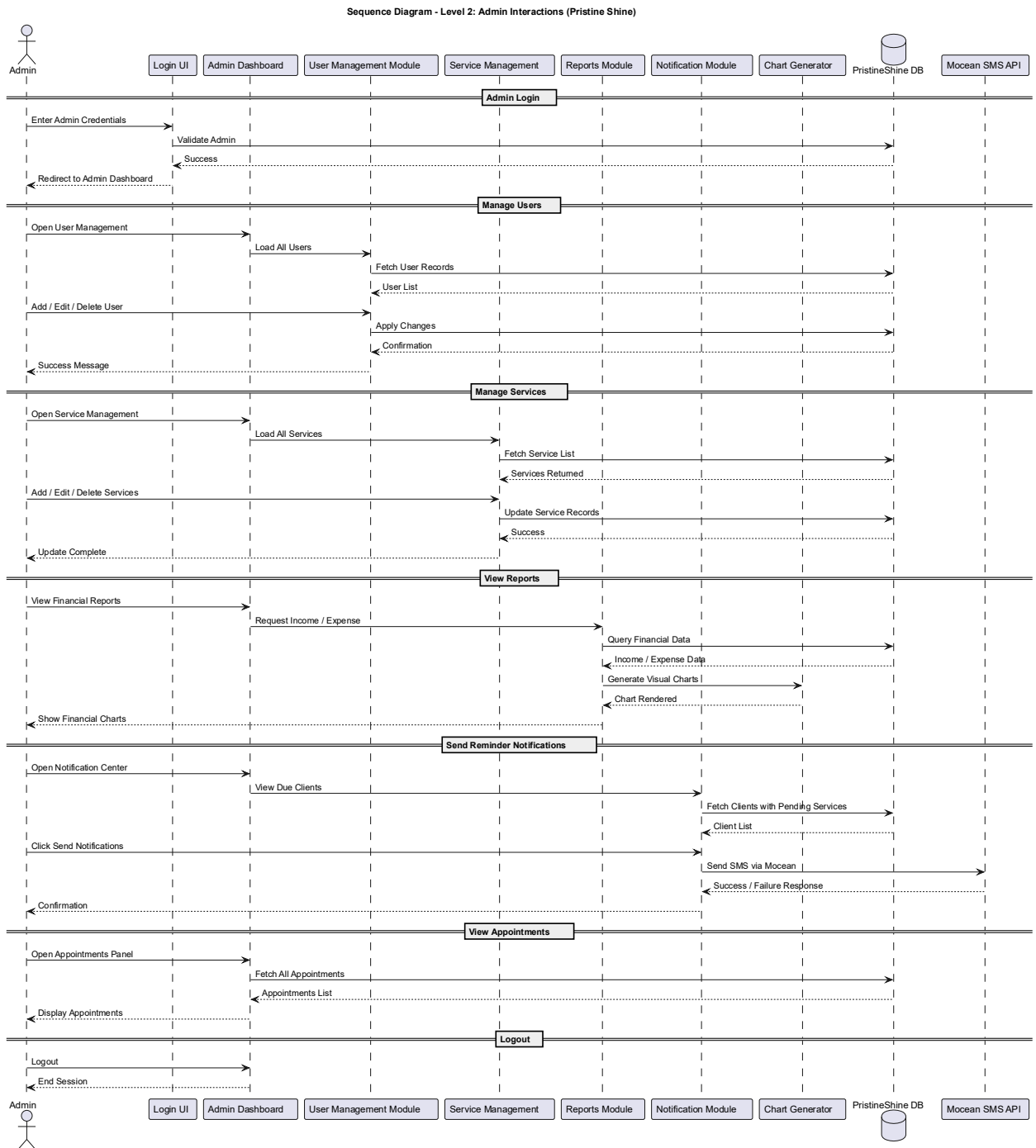


Figure 26 Sequence Diagram -Admin Interactions

5.8 Software Architectural Design

The system follows a **modular architecture** based on a **3-tier design**:

- **Presentation Layer:** WinForms UI using Guna.UI2
- **Logic Layer:** C# classes and helper methods (e.g., Admin Helper, Common Helper)
- **Data Layer:** MySQL Database (local WAMP server), accessed via MySql.Data.MySqlClient

Additional Design Considerations:

- **Loose coupling** via helper classes
- **Session-based authentication**
- **Python AI integration** through external call
- **Mocean API** for SMS gateway
- **Charts** rendered via Guna.Charts.WinForms

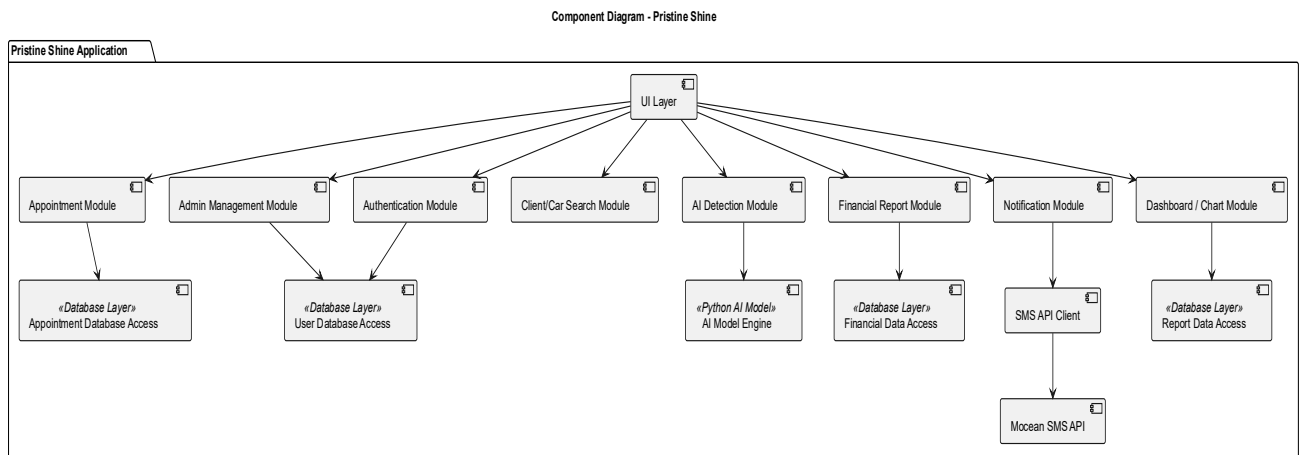


Figure 27 Component Diagram

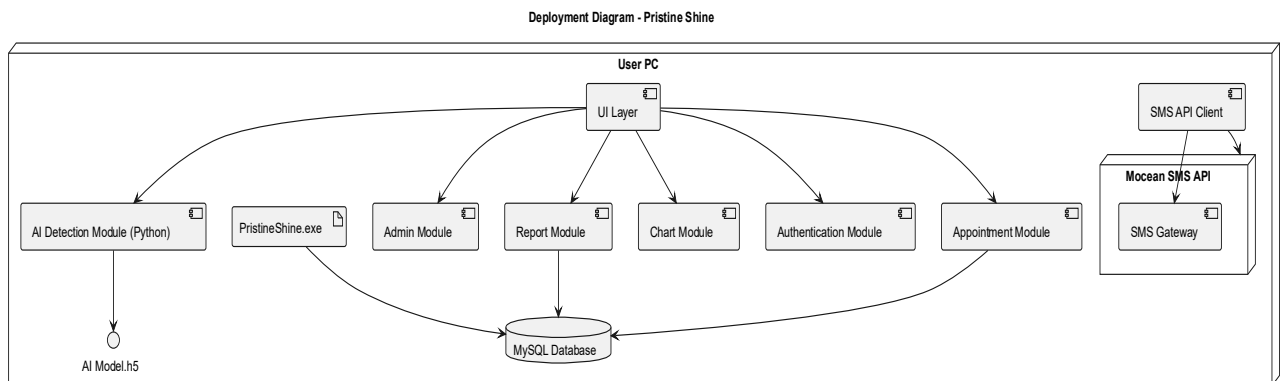


Figure 28 Deployment Diagram

CHAPTER 6: DEVELOPMENT

6.1 Selection of Tools & Technology

The development of the Pristine Shine Car Detailing Management System required a comprehensive technology stack combining desktop application development, database integration, artificial intelligence, and external API communication. The selected tools and technologies were chosen based on project requirements such as offline access, rich user interface, AI-powered damage detection, and automated client communication through SMS.

6.1.1 Hardware Used

The following hardware was used during the development, testing, and deployment phases of the project:

Component	Specification
Processor	Intel Core i5 (4th Gen or above)
RAM	Minimum 4 GB DDR4 8 GB Recommended
Storage	256 GB SSD (preferred for faster performance)
Graphics	Integrated Graphics (Intel UHD / Iris)
Monitor	Full HD (1920x1080)
Operating System	Windows 10 / 11 (64-bit)
Input Devices	Keyboard and Mouse
Internet	Required for setup, updates, and SMS API

Table 12 Hardware Used

6.1.2 Software Used

1. Programming Languages and Frameworks

Layer / Purpose	Technology / Language
Frontend (UI)	C# (Object-Oriented Programming)
Database Layer	MySQL (Relational Database)
AI Module	Python 3.11 (for damage detection)
External Communication	HTTP via Mocean SMS API
Installer	Inno Setup Compiler

Table 13 Software Used Programming Languages and Frameworks

2. IDEs and Tools

Tool / IDE	Purpose
Visual Studio 2022	Core development in C#, UI design, .NET Forms
MySQL Workbench	Designing and managing MySQL databases
VS	Python scripting and debugging
Postman	API testing (Mocean SMS)
Inno Setup	Creating offline installer for deployment
Draw.io / PlantUML	UML and system design diagrams
Snipping Tool	Capturing screenshots of UI and reports

Table 14 Software Used IDEs and Tools

Chapter 7: SOFTWARE TESTING

7.1 Software Testing

Software testing plays a crucial role in the development of the Pristine Shine desktop application. It ensures that the system meets the functional and non-functional requirements set by stakeholders, and that it performs reliably under different conditions. Since the application involves client appointments, financial data, AI-based damage detection, and SMS communication, even minor bugs can lead to incorrect data handling or user dissatisfaction. Therefore, rigorous testing was essential to ensure accuracy, stability, and security of the system.

7.2 Testing Process

Our testing process followed a Black Box Testing methodology since the testers were not required to know the internal code structure. We tested the system functionalities based on the inputs and expected outputs. Testing was performed module-by-module using the following steps:

Requirement Analysis – Functional requirements were identified from use case diagrams and system design.

- **Test Planning** – Defined test scenarios and expected results.
- **Test Execution** – Ran test cases for each module individually.
- **Bug Reporting & Fixing** – Logged issues and returned to development for fixes.
- **Retesting** – Re-ran cases to confirm stability and correctness.

Tools Used:

1. Manual testing:

- MySQL Workbench (for backend testing)
- Visual Studio Debugger
- Console logs for AI Python module
- UI verification using sample inputs

7.3 Black Box Testing

We applied Black Box Testing to test the external behavior of the system by inputting different combinations and observing the outputs, without touching the source code. This was especially important for:

- Login system (valid vs invalid credentials)
- Appointment booking (valid client data vs missing fields)
- SMS sending via Mocean API
- AI damage detection on sample images

7.4 Test Cases

Test Case ID	Test Scenario	Input	Expected Output	Status
TC001	Admin Login	Valid email & password	Redirect to Admin Dashboard	Pass
TC002	Employee Login	Invalid credentials	Show “Invalid Credentials” message	Pass
TC003	Add New User (Admin)	Complete user form	User saved to database	Pass
TC004	Add New User (Admin)	Leave required fields empty	Show validation error	Pass
TC005	Book Appointment (Employee)	Valid appointment details	Appointment stored in database	Pass
TC006	Book Appointment (Employee)	Missing client or car info	Show error message	Pass
TC007	Search Car	Car number or CNIC	Show matching client and appointment data	Pass
TC008	AI Damage Detection	Upload car image	Show detected damage on screen	Pass
TC009	Send Reminder Notification (Admin)	Due appointments present	SMS sent via Mocean API	Pass
TC010	View Reports	Navigate to Reports tab	Show income, expenses,	Pass
TC011	Password Reset (Forgot Password flow)	Valid email & verification code	Allow password reset	Pass
TC012	Password Reset	Invalid code or expired	Show “Invalid or expired code” message	Pass
TC013	Log Out	Click logout	Session cleared and return to login screen	Pass
TC014	Install Software	Run Inno Setup Installer	App, DB, Python, .NET dependencies installed	Pass
TC015	Financial Report Generation (Admin)	Generate button clicked	Report saved/printed	Pass

Table 15 Test Cases

Chapter 8: IMPLEMENTATION AND TRAINING

8.1 Implementation

The implementation of the Pristine Shine Management System followed a structured and modular approach. The software was developed as a Windows desktop application using C# (.NET Framework) and MySQL as the backend. Python was integrated for AI-based image damage detection, and the Mocean API was used for automated SMS reminders.

The application was packaged into a full offline installer to simplify setup for end-users. The installer includes all required components and creates a complete working environment on the user's PC. The system was implemented and tested in a staged approach:

Phase 1: UI/UX Design

Created with Guna UI2 for a modern and responsive user interface.

Phase 2: Admin Panel

Features included: Manage Users, View Reports, Manage Services, View Appointments, and Send Reminders.

Phase 3: Employee Panel

Features included: Search Car/Client, Book & Complete Appointments, AI Damage Detection.

Phase 4: AI & API Integration

Local Python model integration for damage detection.

Mocean SMS API integration for service notifications.

Phase 5: Deployment

Created an offline installer using Inno Setup.

Final database and configurations were bundled.

8.2 Project Overview (Key Features)

The following are the key features of the Pristine Shine system:

- Role-Based Login System (Admin & Employee)
- Client and Car Search by CNIC or Car Number
- Book, Edit, and Complete Car Appointments
- AI Damage Detection via Uploaded or Captured Images
- Dynamic Financial Reports and Chart Visualizations
- Better tracking of appointments, services, and income.
- Reduction in manual effort through digital records.
- Enhanced customer engagement through SMS reminders.

- Improved accuracy with AI-powered damage detection.
- The expected outcome is a professional-grade, maintainable, and efficient software tailored specifically for a car detailing studio environment.
- Admin-Controlled User and Service Management
- Automated SMS Reminders for Due Services (via Mocean API)
- Income Logging and Discount Handling
- Modern Interface using Guna UI and Guna Charts

Expected Outcome:

The Pristine Shine software automates and simplifies the day-to-day operations of a car detailing studio. It ensures:

- Better tracking of appointments, services, and income.
- Reduction in manual effort through digital records.
- Enhanced customer engagement through SMS reminders.
- Improved accuracy with AI-powered damage detection.
- The expected outcome is a professional-grade, maintainable, and efficient software tailored specifically for a car detailing studio environment.

8.3 Training

Skillset Required to Operate the Software:

- **User Role:** Skills Required
- **Admin:** Basic computer knowledge, ability to manage data, interpret reports, and use visual dashboards.
- **Employee:** Ability to navigate the UI, search and update records, use camera/image upload for AI detection.

Training Approach:

- Initial Training Session (1–2 hours):
- Walkthrough of login, dashboard, and core features.
- Explanation of booking and completing appointments.
- SMS sending and report viewing.

Hands-On Practice:

- Use test data to practice service entries and edits.
- Simulate AI detection via image uploads.

Support Material:

- README with step-by-step setup instructions.
- Tooltips and guided prompts within the UI.
- Admin-only module to reset user passwords or assist with forgotten credentials.

Conclusion:

The Pristine Shine Car Detailing Studio Management System is a robust, all-in-one solution for managing clients, services, employees, and financial records. The integration of modern UI frameworks, powerful reporting via charts, AI-based damage detection, and Mocean-powered SMS communication makes it a feature-rich and reliable system.

It has been designed for offline deployment, removing dependency on internet services (except SMS). The software boosts productivity, reduces operational errors, and enhances customer communication—transforming how a car detailing studio operates.

The project demonstrates a successful integration of software engineering principles, machine learning, API communication, and real-world application of database management.

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